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Applying machine learning to a nonlinear spectral mixing model for mapping lunar soils composition using CHANDRAYAAN-1 M^3 data

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ABSTRACT

We present a newly developed method which combines the nonlinear spectral mixing model of Shkuratov et al. (1999) with a machine learning algorithm to map the lunar regolith composition using spectral data. The new method performs orders of magnitude faster than the traditionally used numerical optimization approaches, allowing the mapping of regolith properties (including mineralogical composition, average grain size and optical maturity) over large areas of the lunar surface. A new set of basic mineral spectra of the lunar soil for using with spectral mixing models is proposed. Used together with the nonlinear mixing model (Shkuratov et al., 1999), the set is able to describes Chandrayaan-1 M^3 instrument spectra collected from test areas which includes the Shapley crater with its surroundings containing mare and highland terrains well. The new set includes a virtual "gray component" with a "flat" (constant) spectrum, accounting for the factors that change general surface albedo, such as spectrally neutral components (e.g., agglutinate glasses), errors in the photometric reduction, uncertainties in estimations of lunar regolith porosity q and the mean grain size S of the basic minerals. The proposed new method takes into account the influence of space weathering and nonlinear correlation between the compositional and spectral parameters of the lunar soils delivering values for the optical properties and mineralogical abundance determination of the lunar regolith which are compatible with the results found from lunar samples measurements in the laboratory. The proposed approach can be used for analyzing spectral observations not only of the lunar surface but also for other surfaces with are covered by regolith.

1. Introduction

The energy distribution in a reflectance spectrum in the visible and the Near-Infrared (NIR) range is defined mainly by the chemical/mineralogical composition of the light-scattering layer of the lunar surface. A determination of the lunar regolith composition through spectral observations is in principle simple to achieve. It is possible to analyze the spectra using an optical model to infer the mineralogical composition of the light scattering surface. However, the majority of observed lunar spectra are relatively featureless (see, for instance, Figs. 1 and 2). This fact has the advantage that one can carry out observation with relatively low spectral resolution and is able to replace spectrophotometry by colorimetry (observations in selected spectral bands). However, one has to take into account the drawback that the results of unmixing the spectra lead to various ambiguities in the determined parameters. The situation is aggravated by the fact that the observed reflectance spectra produced by a mixing of the basic chemical/mineralogical components depend on the optical and physical properties of the medium's individual components in a rather complex nonlinear manner.

Traditionally, the simplest approach to determine the lunar soil composition has been to use the linear correlation technique, which relies on direct statistical links between the spectral characteristics and the chemical/mineral composition of lunar soils. Such an approach was followed by the Lunar Soil Characterization Consortium (LSCC) (Pieters et al., 2000, 2002, 2006; Taylor et al., 2001a, 2001b). People using the data from LSCC with the above mentioned technique have assumed that the mineralogical/chemical composition, optical maturity and grain size of the lunar soils linearly correlates with the spectral parameters obtained from photocolorimetric observations (e.g., Clementine CSR

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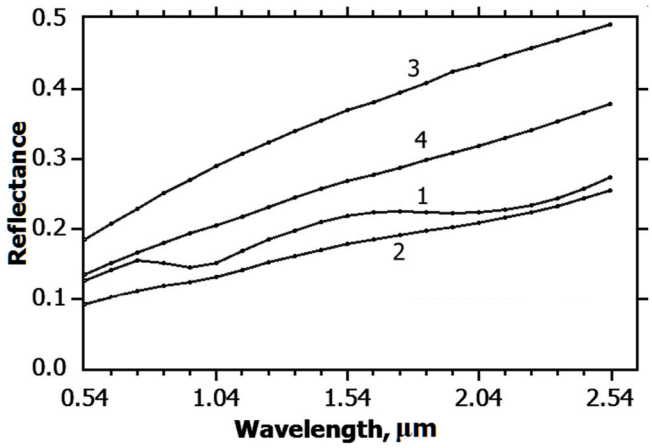

Fig. 1. Typical Chandrayaan-1 M^3 bidirectional reflectance spectra of the lunar surface for test points near the crater Shapley (see Fig. 3) in the RELAB photometric system.

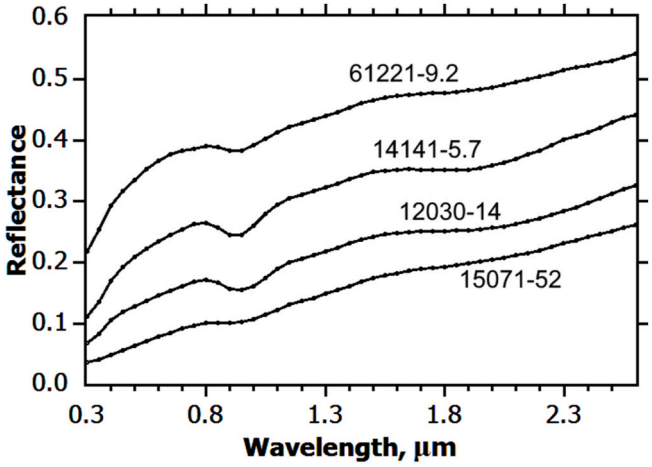

Fig. 2. Bidirectional reflectance spectra $A(\lambda)$ of some LSCC samples (bulk).

UVVIS (Nozette and the Clementine Team, 1994). This correlation can be found using multiple linear regression (MLR) models as proposed in the series of studies by Shkuratov et al. (2003a, 2003b, 2005a; 2005b, 2007) and Pieters et al. (2006).

Alternatively, a nonlinear dependence can be used. One of the widely used nonlinear approaches uses a method proposed by Lucey et al. (1995, 1998, 2000a, b, 2004), based on results found by studying the statistical correlation between laboratory measured optical and composition parameters of lunar LSCC samples using the correlation diagrams A(750 nm) – C(950/750 nm) and A(750 nm) – C(415/750 nm), where $A(\lambda)$ is the albedo at a given wavelength λ and $C(\lambda1/\lambda2) = A(\lambda1)/A(\lambda2)$ is the color index. A more general nonlinear approach was suggested by Korokhin et al. (2008). It uses a kind of artificial neural network – multilayer perceptron (Rumelhart and McClelland, 1986) to find correlations between compositional and optical parameters without any restrictions on the functional behavior of these relations. In a review of nonlinear hyperspectral unmixing techniques, Heylen et al. (2014) point out that the ANN technique is useful in planetary remote-sensing applications to retrieve abundance proportions for a set of basic components (endmembers) from spectral data.

Correlation algorithms are very simple to implement and use, explaining their wide adoption to infer the chemical/mineralogical composition of lunar soils from data of many multispectral cameras, e.g., Clementine UVVIS (Shkuratov et al., 2003b, 2005a, 2005b, 2007; Pieters et al., 2006; Lucey et al., 1995, 1998, 2000a, b; 2004; Korokhin et al., 2008), SMART-1 AMIE (Shkuratov et al., 2003a), SELENE/Kaguya MI (Multiband Imager) (Shkuratov et al., 2018), Lunar Reconnaissance Orbiter LROC WAC (Wide-Angle Camera) (Korokhin et al., 2016; Sato et al., 2017). However, careful attention should be paid to the interpretation of the results found through correlation methods. Firstly, not all optical parameters of a soil are linearly determined by its mineralogic composition. For instance, the content of TiO_2 in lunar soils has a complex nonlinear dependence on its spectral characteristics (Korokhin et al., 2008). In addition, while using a correlation algorithm, one should always remember that the above-mentioned correlation methods are based on a very small lunar set (only several tens) of samples acquired mainly in mare areas. Thus, the representativeness of this set for the entire Moon's surface has to be questioned.

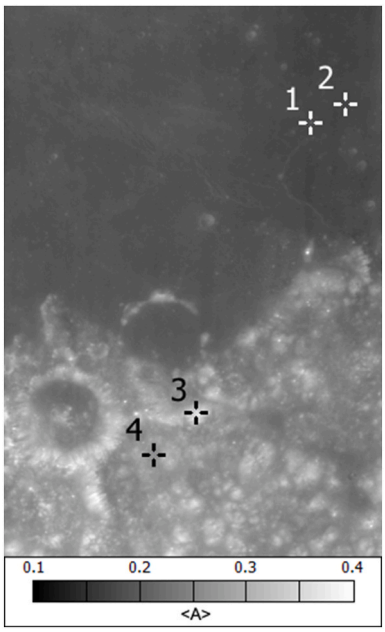

Fig. 3. Mean albedo map of the test area near the crater Shapley. The map was obtained by averaging the Chandrayaan-1 M^3 images in the spectral range 0.54 … 2.60 μm and reducing it to the RELAB photometric system. Four test points, marked by crosses, represent image locations where test spectra for different surface types have been investigated (see Sections 4.2, 5.4 and 5.5).

Another group of approaches does not make any use of correlation relations between optical and composition parameters found from laboratory measurements. These approaches include: the Principal Component Analysis (PCA) method (Johnson et al., 1985), the Modified Gaussian Model (MGM) (Sunshine et al., 1990), the automated band detection technique based on wavelet filtering (Gendrin and Erard, 2003), or the analysis of the absorption band depth (BD) specific for a particular mineral (Shkuratov et al., 2019; Surkov et al., 2020, 2021). All of these methods are based on the analysis of positions and strengths of absorption bands in the spectra and require minimal a priori information. However, these methods have also their own limitations (see, e.g., Poulet and Erard, 2004). They are very sensitive to the effects of particle size and composition, and produce large errors for the quantitative analysis of multicomponent mixtures in cases when several components do not show pronounced spectral features (Moroz and Arnold, 1999). Previous studies applied these methods to very specific mixtures (primarily of olivine and pyroxene) or analyzed the content of a single mineral (e.g., Surkov et al., 2020, 2022).

A more general approach uses a physical-driven light scattering model which is able to consider a number of compositional, structural and optical parameters of the regolith, e.g., optical constants, the particle packing density and sizes, spectral behavior of all end-members, including spectrally neutral components. Among the light scattering

models, the Hapke (1981) and the Shkuratov (Shkuratov et al., 1999) models have found the widest application. The Hapke model was used both for the analysis of laboratory measured spectra of natural or man-made mixtures (Mustard and Pieters, 1987, 1989; Hiroi and Pieters, 1994), and for the study of the mineralogical composition of the lunar regolith using data from orbital observations (for instance, Lucey et al., 2014; Lemelin et al., 2013, 2015, 2018). In spite of all the advantages and the simplicity of the Shkuratov model (the detailed comparison between these theories for usage as a spectral mixing model is given in Poulet et al., 2002; Shkuratov et al., 2012, 2013), the Shkuratov model was most often used only for analyses of laboratory spectra of natural or man-made mixtures (for instance, Poulet and Erard, 2004), and rarely for lunar surface observations. In this paper, we make one of the first, and as we think, successful attempts to apply the Shkuratov et al. (1999) nonlinear mixing model to map the compositional and physical parameters of the lunar surface. For this we use the spectral observations of the hyperspectral imager M^3 (Pieters, 2009), which operated onboard the Indian Chandrayaan-1 lunar spacecraft (Goswami and Annadurai, 2009).

In the first part of this paper, we summarize some of the key points of the Shkuratov et al. (1999) spectral mixing model. Then we discuss how to choose a set of basic minerals to describe most of the features of the lunar spectra and carry out tests with this set to model Chandrayaan-1 M^3 data. After this, we describe a new method based on machine learning that allows the Shkuratov spectral mixing model to be used to map the lunar soils properties (composition, grain size and maturity) for big datasets, i.e. large lunar areas. Using the improved Chandrayaan-1 M^3 spectra, we then perform various tests of our new method in an area near the crater Shapley, which contains both maria and highland areas.

2. Method to determine the lunar soils composition and its practical implementation

2.1. The idea of the method

The essence of the method is to fit a modeled reflectance spectrum $A_M(\lambda)$ of a mineral mixture to an experimentally measured (observed) reflectance spectrum $A_O(\lambda)$ by varying the parameters of the mixture (concentrations of basic components, its grain sizes, packing density, optical constants, etc.). The modeled spectrum $A_M(\lambda)$ is calculated using laboratory measured spectra of basic components (minerals) $A_S(\lambda, j)$ (j is a basic sample index) and given parameters of the mixture.

In this work, we use a nonlinear mixing model based on the Shkuratov's theory of spectral albedo of particulate surfaces (Shkuratov et al., 1999). One of the first practical applications of this model for quantitative analysis of laboratory mineral mixtures was given by Poulet and Erard (2004). Our proposed algorithm retrieves the concentrations of basic elements (minerals) and physical characteristics (packing density and grain size) for the spectrally observed surface. For a brief description of the nonlinear mixing model and formulae, see the Appendix.

2.2. Algorithm of the modeling

The practical implementation of the method can be summarized in the following steps.

1. Calculate spectra of extinction coefficients $k(\lambda, j)$ (Eq. (A1)) (or optical density $\tau(\lambda, j)$ (Eq. (A2)) for nontransparent components) from laboratory bidirectional reflectance spectra $A_S(\lambda, j)$ for each basic element j, using laboratory measured parameters of these basic samples: real part of a refractive index n, an average grain size S, and a packing density q (a volume fraction filled by the particles).
2. Calculate albedo of a particle in backward and forward directions $r_b(\lambda, j)$ and $r_f(\lambda, j)$ respectively for each basic component using Eq. (A8). For transparent components (clinopyroxene, orthopyroxene, olivine, plagioclase), the optical density τ is calculated using formula (A9) with the parameter $S(j)$ for the modeled medium (mixture) considering effects of the optical maturity (Eqs. A13 – A16). For nontransparent components (for instance, ilmenite and agglutinates), the optical layer density τ is inferred from Eq. (A2).
3. Calculate bidirectional reflectance spectra $A_M(\lambda)$ of the mixture (multicomponent medium/surface) using Eq. (A12) with parameters (packing density q, reduced nanophase iron concentration c_{nFe}, concentration of mineral $c(j)$, and albedo of particles $r_b(\lambda, j)$, $r_f(\lambda, j)$ for each mineralogic component of the mixture), which provide the best approximation of experimentally measured (observed) spectrum $A_O(\lambda)$. These free parameters q, c_{nFe}, $c(j)$ and $S(j)$ can be obtained, for instance, using an automatic best-fit procedure (see below).

We emphasize the difference in the meanings of the parameters q and S for the step (1) and steps (2)–(3). In the first case, they are laboratory-measured parameters of the basic samples, while in the second case they are fitted values for a modeled mixture (medium).

Thus, the algorithm allows to retrieve the concentrations of the basic elements $c(j)$ and the physical characteristics (packing density q, maturity c_{nFe} and grain size S) of the spectrally observed surface.

2.3. Applying the numerical optimization procedure to determine the parameters of a mixture

We started this work using an optimization procedure to determine the parameters of the mixture. The root mean square deviation (RMSD) was used as an objective function

$$\sigma = \sqrt{\frac{1}{N}\sum \left(A_M(\lambda) - A_O(\lambda)\right)^2} \quad (1)$$

where $A_M(\lambda)$ is the modeled spectrum, and $A_O(\lambda)$ is the observed one; the summation is performed over all wavelengths (spectral channels) in the spectrum, N denotes the number of points in the spectrum array.

To find the mixture parameters, we used the Nelder and Mead (1965) optimization method (so-called the modified simplex method). We have modified the method to have the possibility to set an initial simplex with different initial facet lengths (initial steps). The success of a multi-parameter optimization depends significantly on the choice of the initial fit parameters and the initial facet lengths, because the function (Eq. (1)) depends on a large number of free parameters (8 in our case) and has many local minima. All free parameters are normalized to a uniform range applying corresponding scaling factors.

Our tests of the algorithm have shown that finding the true solution (global minimum of Eq. (1)) depends in a very complex way on the chosen initial values. The method is especially sensitive to initial values of the grain size S_0. Often the method finds one of the local minima and does not provide the optimal solution.

To resolve this problem, we used an exhaustive search to find the optimal initial value for the grain size S_0, carrying out the optimization with a small initial step (simplex facet) of S. During every iteration, the algorithm searches for a solution in a narrow "corridor" of S. This ensures that the algorithm does not miss the global minimum. It is necessary to choose optimal parameters for the exhaustive search: too small an initial step ΔS slows down the calculations, too large ΔS can lead to missing the optimal solution. For our implementation, the best results were obtained for the search range 1 μm $\leq S_0 \leq$ 100 μm with a step of 1 μm, and an initial simplex facet $\Delta S = 1$ μm. For young lunar areas, the upper limit is shifted to larger values, up to 500 μm.

2.4. Testing the algorithm with man-made mixtures

Before being applied to lunar spectra, the algorithm had been tested on the spectra of man-made mixtures with well-known parameters. Fortunately, the RELAB Spectral Database (https://sites.brown.edu/re

lab/) contains a series of such measurements. We have selected 3 mixtures.

1) 2-component mixture (50%: 50%) MS-CMP-016 (spectrum C1MS16) of olivine MS-CMP-014 (C1MS14) and pyroxene MS-CMP-015 (C1MS15) (Fig. 4);
2) 3-component mixture (33%: 33%: 33%) XT-CMP-036 (C1XT36) of olivine PO-CMP-081-C (C2PO81), orthopyroxene PE-CMP-041-C (C1PE41) and anorthite PA-CMP-060-C (C1PA60) (Fig. 5);
3) 3-component mixture (40%: 30%: 30%) SC-EAC-007 (C1SC07) of olivine SC-EAC-002 (C1SC02), low calcium pyroxene SC-EAC-009 (C1SC09) and Odessa Meteorite metal powder SC-EAC-006 (C1SC06) (Fig. 6).

Mixtures (1) and (2) were part of the study published in (Hiroi and Pieters, 1992), mixture (3) were prepared by E. A. Cloutis (Cloutis et al., 1990).

To calculate the absorption spectra $k(\lambda)$ (Eq. (A1)), we used grain size values S calculated according to the recommendations given by Pieters et al. (2002) (see Table 1). The Odesa Meteorite metal powder in the mixture #3 was used as a nontransparent component. Since all the samples for these tests are immature, we did not consider space-weathering factors.

The results of our method applied to the man-made mixtures are presented in Table 1 and Fig. 4b, 5b and 6b. They demonstrate that the algorithm provides a good approximation to the laboratory spectra (errors are in the 1.0–1.6% range). Parameters of the man-made mixtures (grain size and components proportion) can be restored with high enough accuracy.

Note, that although the Shkuratov model was initially designed for transparent particles, it works properly with nontransparent particles as well, as becomes evident from the test for the mixture #3 (the application of the model to nontransparent components is described in Section 2.2 and Appendix A.2).

3. Basic spectra set for spectral mixing

3.1. Selecting the representative spectra set of base components (minerals) for spectral mixing

Having an adequate set of laboratory spectra representing the basic components (end-members) of the lunar soils is essential to be able to correctly determine the soil's composition from spectrometric observation. Various spectral end-member sets have in the past been proposed for spectral (un)mixing. We mention here only two popular sets: the set of Cloutis and Gaffey (1993) (Fig. 7 and Table 2) and the set of Noble et al. (2006) (Fig. 8 and Table 3). The end-member set proposed by Cloutis and Gaffey (1993) contains 5 components (clinopyroxene – CPX, orthopyroxene – OPX, olivine – OLV, plagioclase – PLG and ilmenite – ILM) of terrestrial (TE) origin and is homogeneous in grain size. The set of Noble et al. (2006) additionally includes agglutinates (AGG) and glass (GLS); the components are both of terrestrial (TE) and lunar (LU) origin and differ significantly in grain size. As we can see from Figs. 7 and 8, minerals of the same type have different spectra in these two sets. This can be explained by the difference in grain size and the origin of the measured samples.

We note that ilmenite had not been taken widely into account during remote mineralogic studies of the Moon's surface. Primarily because it was considered 'featureless' in the VIS-NIR spectral range (see the spectra of the basic minerals in Figs. 7 and 8) having only one spectral characteristic – the extremely low reflectance. This property makes ilmenite indistinguishable from agglutinates in remote spectrophotometric studies (Fig. 8). However, theory predicts that ilmenite should have a band near 1.5 μm caused by the Fe–Ti intervalence charge transfer transition (Burns, 1993). Such bands, unlike ones produced by the crystal field, can be quite shallow and broad, and can be unresolvable or superimposed on other absorption bands (in case of ilmenite in mixtures even at significant abundances). Recent laboratory and remote sensing investigations have demonstrated the possibility to identify the band near 1.5 μm and the importance to consider an ilmenite presence (e.g. Isaacson et al., 2011; Lemelin et al., 2013; Surkov et al., 2020; Robertson et al., 2022). In this work, we use the recent spectral library form the Lunar Rock and Mineral Characterization Consortium (LRMCC, Isaacson et al., 2011). The measured reflectance spectra of isolated ilmenite from the LRMCC spectral library show a proliferative reflectance increase after 1.5 μm that we interpret as an absorption feature (Fig. 9a).

Various attempts to use these two sets (separately and in different combinations) as a basis for spectral (un)mixing did not deliver qualitatively and quantitatively satisfactory results.

Therefore, we created a new set of basic spectra. The following empirical criteria were used to identify the new spectra.

1) The new set should provide a minimal deviation between modeled and observed Chandrayaan-1 M^3 spectra;
2) Physically acceptable values of the modeled mixtures parameters (concentrations of basic elements, grain size, maturity, etc.) have to result from a use of these spectra.;
3) The spectra have to originate, if possible, from samples of lunar origin;

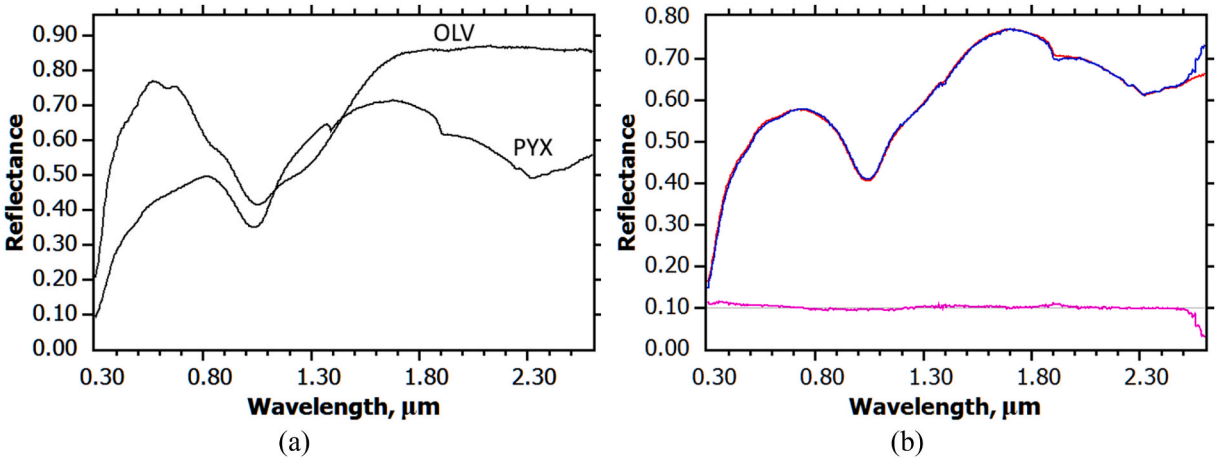

Fig. 4. RELAB bidirectional reflectance spectra of pure minerals (a), and their 50%: 50% mixture MS-CMP-016 (spectrum C1MS16) (b, blue). Calculated spectrum (b, red) and difference between calculated and observed spectrum + 0.1 (b, magenta). OLV – olivine MS-CMP-014 (C1MS14), PYX – pyroxene MS-CMP-015 (C1MS15).

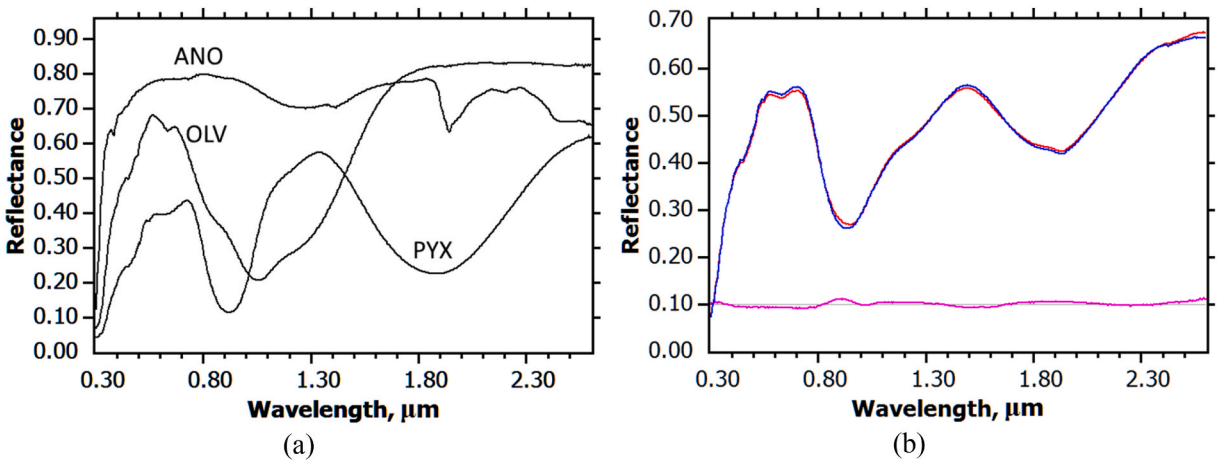

Fig. 5. RELAB bidirectional reflectance spectra of pure minerals (a), and their 33%: 33%: 33% mixture XT-CMP-036 (spectrum C1XT36) (b, blue). Calculated spectrum (b, red) and difference between calculated and observed spectrum + 0.1 (b, magenta). OLV – olivine PO-CMP-081-C (C2PO81), PYX – orthopyroxene PE-CMP-041-C (C1PE41), ANO – anorthite PA-CMP-060-C (C1PA60).

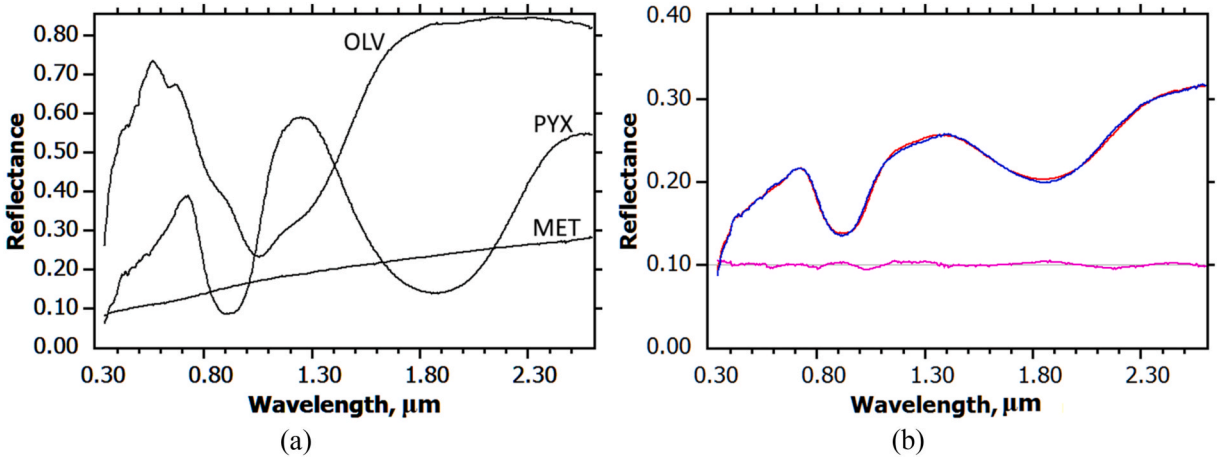

Fig. 6. RELAB bidirectional reflectance spectra of pure minerals and metal powder (a), and their 40%: 30%: 30% mixtures SC-EAC-007 (C1SC07) (b, blue). Calculated spectrum (b, red) and difference between calculated and observed spectrum + 0.1 (b, magenta). OLV – olivine SC-EAC-002 (C1SC02), PYX – low calcium pyroxene SC-EAC-009 (C1SC09), MET – Odessa Meteorite metal powder SC-EAC-006 (C1SC06).

Table 1
Results of applying the method to man-made mixtures. $\sigma_{Rel} = 100\%\ \sigma/<A>$ denotes the relative error of the spectrum approximation, where σ is the standard error, $<A>$ is the albedo averaged over the entire M^3 spectral range (see Fig. 3). OLV – olivine, PYX – pyroxene, ANO – anorthite, MET – Odessa Meteorite metal powder (see Figs. 4–6).

Mixture	Source components proportions	Grain size range, μm	S for Eq. (A1), μm	Restored components proportions	Restored grain size S, μm	σ_{Rel},%
OLV + PYX	50%: 50%	0 … 75	38	56%: 44%	32.9	1.6
OLV + PYX + ANO	33%: 33%: 33%	45 … 75	45	39%: 34%: 27%	43.3	1.1
OLV + PYX + MET	40%: 30%: 30%	45 … 90	45	40%: 23%: 37%	39.4	1.0

4) Spectra from the set should be similar in origin, grain size S and packing density q distribution, and the spectra should have been acquired under comparable measurement conditions.

Please note that knowing the correct values for S and q of laboratory measured samples is very important to calculate useable extinction coefficients spectra $k(\lambda)$ from bidirectional reflectance spectra $A(\lambda)$ (see Appendix). Errors in these parameters lead to errors in the resulting model parameters, first of all in the values of the found concentrations and grain sizes. Unfortunately, none of the above-mentioned studies give the values for the packing density of the basic samples. Therefore, in this work we use as a reasonable value for the lunar soils $q = 0.5$ (this value has been measured both on the lunar surface *in situ* and has been determined through laboratory sample analyses). Equally important is a knowledge of the real part of the refractive index n measured for the minerals isolated from the samples. Unfortunately, this information is also absent in the spectra databases. Therefore, we used $n = 1.6$ for transparent elements, and $n = 2.0$ for nontransparent ones (for instance, ilmenite). Selecting the value of 2.0 is related to the limitations of equations (A2) – (A7) for a wavelength less than 0.5 μm when calculating the optical density τ for ilmenite.

Our resulting set of basic elements (minerals) is presented in Table 4. Bidirectional reflectance $A(\lambda)$ and extinction coefficient $k(\lambda)$ for the new spectra are shown in Fig. 9a and b, respectively. The set is based on the Lunar Rock and Mineral Characterization Consortium (LRMCC) database (Isaacson et al., 2011; https://sites.brown.edu/relab/lunar-rock-

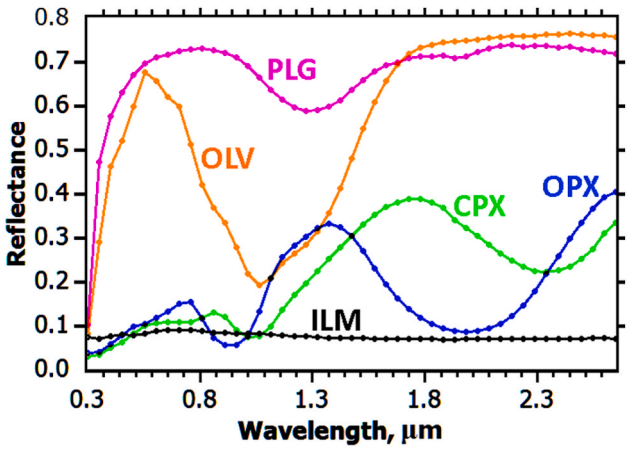

Fig. 7. Spectra set of base elements proposed in (Cloutis and Gaffey, 1993). CPX denotes clinopyroxene, OPX denotes orthopyroxene, OLV denotes olivine, PLG denotes plagioclase PLG and ILM denotes ilmenite.

Table 2
Spectra set of base elements proposed in (Cloutis and Gaffey, 1993). Origination: TE – terrestrial; LU – lunar.

Component	Label	RELAB_ID	RELAB spectrum	Grain size range, μm	Origination
Plagioclase	PLG	SC-EAC-037	C1SC37	45 … 90	TE
Olivine	OLV	SC-EAC-002	CASC02	45 … 90	TE
Orthopyroxene	OPX	SC-EAC-036	C1SC36	45 … 90	TE
Clinopyroxene	CPX	SC-EAC-035	C1SC35	45 … 90	TE
Ilmenite	ILM	SC-EAC-034	C1SC34	45 … 90	TE

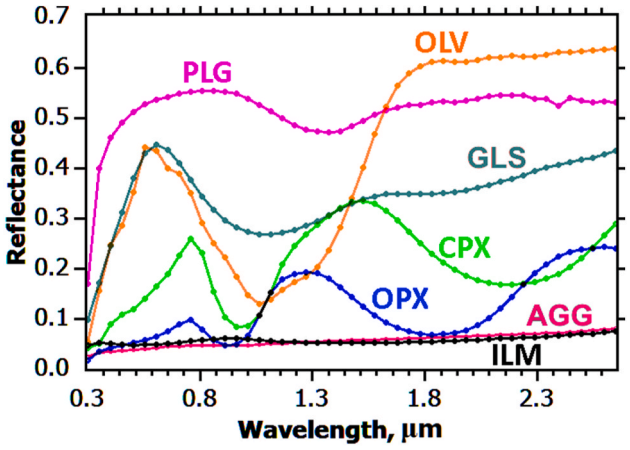

Fig. 8. Spectra set of base elements proposed in (Noble et al., 2006). CPX denotes clinopyroxene, OPX denotes orthopyroxene, OLV denotes olivine, PLG denotes plagioclase PLG, ILM denotes ilmenite, AGG denotes agglutinates and GLS denotes glasses.

and-mineral-characterization-database/) and contains clinopyroxene, orthopyroxene, olivine, plagioclase and ilmenite. We have selected spectra of fine fraction with the same grain size distribution (<45 μm). To calculate the extinction spectra, in Eq. (A1) we use the value of the parameter $S = 30$ μm. This value is close to the mean value of the size distribution range, which has been substantiated by Pieters et al. (2002). Our new basic set has the advantage that it has a completely

Table 3
Spectra set of base elements proposed in (Noble et al., 2006). Origination: TE – terrestrial; LU – lunar.

Component	Label	RELAB_ID	RELAB spectrum	Grain size range, μm	Origination
Agglutinates	AGG	LS-CMP-045	C2LS45	500 … 1000	LU
Clinopyroxene	CPX	LS-CMP-009	C2LS09	0 … 250	LU
Glasses	GLS	LS-CMP-035-A	CALS35	0 … 125	LU
Ilmenite	ILM	JB-JLB-308	CJB308	0 … 125	TE
Olivine	OLV	PO-CMP-026	C2PO26	0 … 125	TE
Orthopyroxene	OPX	SB-RGB-006	C1SB06	0 … 0	TE
Plagioclase	PLG	SW-CMP-012	C1SW12	0 … 500	TE

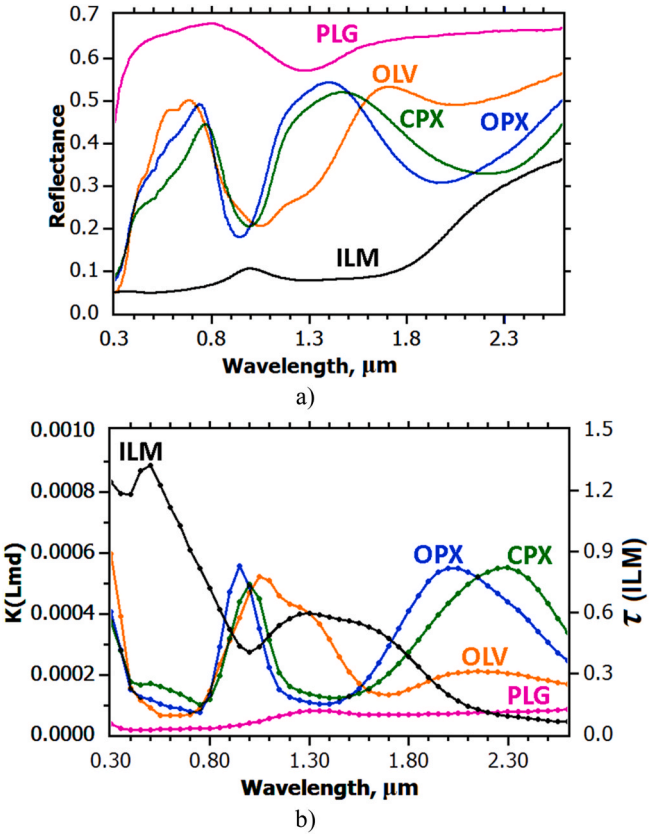

Fig. 9. (a) Bidirectional reflectance spectra $A(\lambda)$, and (b) extinction coefficient $k(\lambda)$ spectra of basic components (minerals) (see Table 4).

homogeneous grain size and was acquired under identical spectral measurement conditions. We want to point out that even if the selected value of $S = 30$ μm in this study should not be correct, that this does not influence the general proportion between the concentrations of the basic elements in the modeled mixtures.

It is also important to keep in mind that is very difficult to form a universal basic set of spectra, which is suitable for the approximation of all soil types of the lunar surface. Our set was specifically selected to describe the M^3 spectra of the lunar areas studied in this work. Please note that the problem of forming a set of basic spectra deserves to be a topic of a separate paper.

Table 4
Basic spectra set used in this work. n – refractive index.

Sample Name	Label	RELAB_ID	Sample Number	Grain size range, μm	S for Eq. (A1), μm	n
Deep Brown Pyroxene	CPX	LR-CMP-215	70017	0 … 45	30	1.6
Light Brown Pyroxene	OPX	LR-CMP-220	70035	0 … 45	30	1.6
Olivine	OLV	LR-CMP-212	15555	0 … 45	30	1.6
Plagioclase	PLG	LR-CMP-217	70017	0 … 45	30	1.6
Ilmenite	ILM	LR-CMP-218	70017	0 … 45	–	2.0

3.2. The "gray component"

As it is seen in Fig. 9 and Table 4 for our basic spectra set, we selected only spectra of minerals with pronounced spectral features: clinopyroxene, orthopyroxene, olivine, plagioclase and ilmenite. Tests have shown that only such basic components can be detected spectroscopically. However, beside these minerals, the real lunar soils contain other elements, for example, agglutinate glasses, which are spectrally rather neutral. These components cannot be recognized spectrally, but they affect the overall albedo.

We propose to add to the basic set a virtual "gray component" with "flat" (constant) spectrum. This component takes into account the influence of factors changing the general albedo, such as spectrally neutral components (e.g., agglutinate glasses and other non-crystalline parts of the lunar regolith darkened by space-weathering agents), errors in the photometric reduction and incorrect value of the porosity q, the real part of refractive index n and grain size S for the basic minerals. We note that in various efforts to map the lunar mineralogy several spectral unmixing models including synthetic spectrally neutral components were previously proposed and tested. Combe et al. (2008) introduced two endmembers: a positive slope spectrum that represents surface maturity, and a positive constant across the spectrum that compensates for different grain sizes and illumination variations. Unmixing results for all possible linear combinations of weighted endmembers spectra were studied and the best fit was retained; no radiative transfer model was employed. Recent analysis of *in-situ* spectral measurements of glassy and non-glassy materials of lunar regolith by the Yutu-2 rover (Chang'E−4 mission) was also performed using spectrally flat and featureless endmember spectra of agglutinate glass (Separate from LUNA 20) and carbonaceous chondrites (Yang et al., 2022). Note that Yang et al. (2022) employed the Hapke radiative transfer model (Hapke, 1981) for spectral unmixing. The same Hapke spectral unmixing model was used for retrieving mineral abundances and grain sizes of the regolith and rocks at the Chang'E−5 landing site (Lin et al., 2022); the green/orange glasses and black beads form RELAB collection were used as endmembers representing quenched volcanic glasses.

Notwithstanding the substantial progress in spectral unmixing modeling, the presence of glassy materials with varying chemical compositions and complex structures further complicate mineral abundance retrievals (Hu et al., 2023). The main advantage of our method, i.e. the "gray component" usage, is to allow one to omit ambiguities and spectral diversity of glassy and agglutinic materials in the mineral composition retrieving procedure. The content of this component is primarily controlled by the albedo level and not by the absorption spectra features.

The gray component is used in the following manner. Using Eqs. (A2 – A7), the optical density τ_{Gr} of the gray component is calculated with some values of the real part of the refractive coefficient n and albedo A_{Gr}. To make the gray component nontransparent enough, $n = 2.0$ is used. The value of A_{Gr} depends on the characteristics of the real lunar surface area. But our tests show that the resulting concentrations of the basic components $c(j)$, maturity c_{nFe} and grain size S depend weakly on the value of A_{Gr} (see Tables 5 and 6). Note that it is possible to modify our algorithm to fit A_{Gr}, too. But even with a fixed value of the gray component's albedo, the algorithm gives stable and feasible concentrations of the basic elements. In this study, we use $A_{Gr} = 0.15$ for Chandrayaan-1 M^3 spectra so that the sum of the concentrations of all base components, including the gray component, is approximately 100%. Note that similar to the other nontransparent components, τ_{Gr} is not modified by the effect of maturity.

Using the above values for τ_{Gr} in Eq. (A8), the albedo of a particles (r_b and r_f) of the gray component is calculated. Finally, these values are used in Eq. (A12) to calculate the spectra of the mixture.

We do not use agglutinate in our basic components set as, for example, is done by Noble et al. (2006). In Fig. 10, the reflectance spectrum of agglutinate LS-CMP-045 used in Noble et al. (2006) is shown. It is very similar to the typical lunar soil spectrum. Compare, for instance, Fig. 10 with the spectrum of sample 12030–14 in Fig. 2.

We can apply our fitting algorithm to these spectra to obtain the composition of these agglutinate samples. The result is given in Table 7. As can be seen from the table, composition, grain size and maturity are similar to the values of typical lunar soils. This can be explained by the fact that agglutinate particles include grains of unmodified basic minerals (Fig. 11), that make a significant contribution to the spectra of this sample.

Including the agglutinate samples like LS-CMP-045 (Noble et al., 2006) in the basic spectra set distorts the results of unmixing, since it is impossible to separate the spectral contribution of the grains of the basic minerals in the studied soil from their inclusions in agglutinates. In Table 8, the results of the application of the fitting procedure to Chandrayaan-1 M^3 spectra with the basic spectra set containing agglutinate LS-CMP-045 instead of the gray component are presented. A direct comparison of Table 8 with 9 shows a significant difference in the concentrations of the basic minerals CPX, OPX, OLV PLG, ILM and the grain size. Moreover, for sites #2 and #4, the concentration of agglutinate is too high (for site #2, it is even higher than 100%). This suggests that in the process of the spectral unmixing, the agglutinate can take the principal role, suppressing the influence of the other basic components.

4. Application of the fitting algorithm to the Chandrayaan-1 M^3 spectra

4.1. Simplifications of the model

Applying the algorithm to observed reflectance spectra of areas with unknow mineralogic composition should provide the following quantities: 5 concentrations $c(j)$ for the basic components (Table 4); 4 grain sizes (average length of light propagation in a particle between internal reflections) $S(j)$ for the transparent basic minerals (clinopyroxene, orthopyroxene, olivine, plagioclase); concentration of the reduced nanophase iron (nFe^0) c_{nFe}; concentration of the gray component c_{Gr}; and packing density of the modeled medium (volume fraction filled by the particles) q. In total, these are 12 free parameters. Having so many free parameters in a numerical optimization problem, can lead to ambiguous solutions. This in turn can complicate the interpretation of the obtained results.

We therefore try to reduce the number of free parameters through the following simplifications.

1) the grain size is assumed to be the same for all basic minerals. This simplification is based on the foundation that all basic lunar minerals have a similar hardness, and the crushing process produces grains of approximately the same size;
2) the packing density of the modeled medium is fixed at $q = 0.5$;
3) the modeled medium (surface) is a mixture of coarse particles of size $\gg \lambda$ (see Appendix A.4);

Table 5
Influence of the "gray component" albedo A_{Gr} on result of the fitting procedure. Test point #1. $\sigma_{Rel} = 100\%\ \sigma/<A>$ is the relative error of the spectrum approximation, where σ is standard error, $<A>$ is the mean albedo (see Fig. 3).

A_{Gr}	τ_{Gr}	c_{Gr}, %	S, μm	c_{nFe}	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	Σc, %	σ_{Rel}, %
0.05	1.304	129.12	137.30	0.086	16.36	6.03	6.44	23.89	6.70	188.54	0.771
0.10	0.441	74.06	137.20	0.086	16.91	6.22	6.66	24.57	6.93	135.35	0.772
0.20	0.180	45.01	137.60	0.086	17.64	6.51	6.97	25.61	7.24	108.98	0.773
0.30	0.095	35.11	137.70	0.086	17.98	6.63	7.11	26.07	7.38	100.28	0.774
0.40	0.053	30.33	137.30	0.086	18.18	6.68	7.20	26.24	7.48	96.10	0.774
0.50	0.030	27.68	138.30	0.086	18.23	6.77	7.22	26.54	7.47	93.91	0.774
Mean			137.57	0.086	17.55	6.47	6.93	25.49	7.20		0.773
StdDev			0.41	0.000	0.76	0.29	0.32	1.04	0.32		0.001

Table 6
Influence of the "gray component" albedo A_{Gr} on result of the fitting procedure. Test point #4. $\sigma_{Rel} = 100\%\ \sigma/<A>$ is the relative error of the spectrum approximation, where σ is the standard error, $<A>$ is the mean albedo (see Fig. 3).

A_{Gr}	τ_{Gr}	c_{Gr}, %	S, μm	c_{nFe}	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	Σc, %	σ_{Rel}, %
0.05	1.304	90.61	41.80	0.071	10.35	1.82	0.00	48.80	11.11	162.69	0.636
0.10	0.441	51.31	39.50	0.071	11.07	1.52	0.00	48.23	11.33	123.46	0.636
0.20	0.180	30.39	43.10	0.071	10.29	2.03	0.00	50.63	11.35	104.70	0.636
0.30	0.095	24.31	42.00	0.073	10.54	2.04	0.00	50.34	11.48	98.71	0.636
0.40	0.053	20.22	41.60	0.071	10.65	2.03	0.15	50.96	11.62	95.61	0.640
0.50	0.030	16.67	43.10	0.071	10.90	1.81	0.00	51.45	11.68	94.19	0.637
Mean			41.85	0.071	10.63	1.88	0.03	50.07	11.43		0.637
StdDev			1.32	0.001	0.31	0.20	0.06	1.27	0.21		0.001

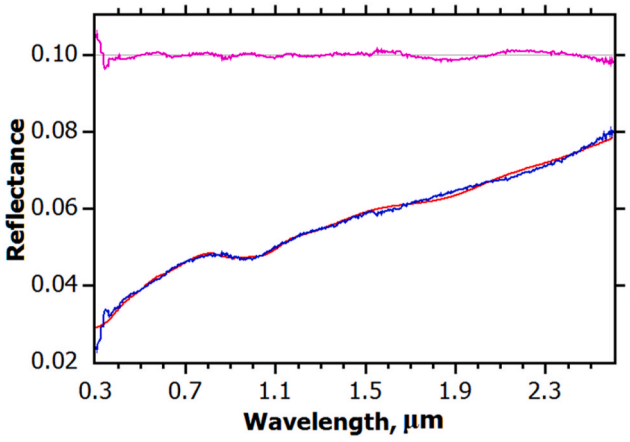

Fig. 10. Best fit for the RELAB lunar agglutinate sample LS-CMP-045 (spectrum C2LS45, d = 500 … 1000 μm), used by Noble et al. (2006) as a basic component. Blue is a measured spectrum; red is a calculated spectrum; magenta is a difference between calculated and measured spectrum +0.1. (See also Table 8).

4) the nFe^0 grains are located in the regolith particles rinds with a thickness of 0.1 μm (see Appendix A.4);
5) the refractive index n for all the transparent basic minerals (clinopyroxene, orthopyroxene, olivine, plagioclase) is equal to 1.6;
6) the refractive index n for nontransparent components (ilmenite, gray component) is equal to 2.0 (as discussed before).

These simplifications reduce the number of free parameters to 8: 5 × $c(j)$, c_{nFe}, c_{Gr} and S.

4.2. Result of applying the fitting algorithm to the M^3 test spectra

To test our algorithm on the Chandrayaan-1 M^3 spectra, we have selected 4 sites near the crater Shapley (see Figs. 1 and 3). Sites #1 and #2 are in Mare Crisium. Sites #3 and #4 are located in the highlands. The used spectra are generated from the original M^3 spectra by applying the technique developed by Shkuratov et al. (2019) and Surkov et al. (2020, 2021). Their technique is based on the Gaussian λ-convolution of spectra and Fourier filtration of the images, which reduce the known strip structure visible in the M^3 images making the spectra suitable for a quantitative analysis. Additionally, we have resampled the M^3 spectra to a uniform wavelength step size of $\Delta\lambda = 0.1$ μm to reduce the size of the processed data. Our experiments have shown that this does not affect the unmixing results, but reduces the computation time.

Before being used the M^3 spectra must be radiometrically reduced to the RELAB photometric system. A direct albedo calibration of the M^3 spectrum obtained from the image M3G20090607T110414, which contained the Apollo-16 landing site, with the RELAB spectrum of the representative Apollo-16 soil sample 62231 was made. The best agreement between the spectra (mean square deviation = 1.4%) is given by the following relation:

$$A_{RELAB} = 1.685\ A_{M3} + 0.0355, \quad (2)$$

where A_{RELAB} is albedo in the RELAB photometric system and A_{M3} is the M^3 albedo.

The above-mentioned procedure is widely used to reduce data from various photometric observations into the RELAB photometric system (e.g., Pieters et al., 1991; McEwen et al., 1998; Pieters, 1999; Shkuratov

Table 7
The results of applying the fitting procedure to the RELAB lunar agglutinate sample LS-CMP-045. $<A>$ is the mean albedo (see Fig. 3), σ is the standard error and $\sigma_{Rel} = 100\%\ \sigma/<A>$ is the relative error of the spectrum approximation.

Sample	S, μm	c_{nFe}	c_{Gr}, %	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	Σc, %	σ	σ_{Rel}, %
LS-CMP-045	57.1	0.052	17.29	7.91	0.00	3.54	6.09	3.19	38.02	0.00088	1.55

Table 8
The results of applying the fitting procedure to Chandrayaan-1 M^3 spectra with base spectra set containing agglutinate LS-CMP-045 instead of the gray component. $<A>$ is the mean albedo (see Fig. 3), σ is the standard error and $\sigma_{Rel} = 100\%\ \sigma/<A>$ is the relative error of the spectrum approximation.

Site	S, μm	c_{nFe}	c_{AGG}, %	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	Σc, %	σ	σ_{Rel}, %
1	87.7	0.033	65.53	8.53	8.56	3.55	40.50	0.00	118.11	0.00186	0.93
2	49.9	0.139	109.69	0.00	4.47	0.44	30.86	2.04	147.50	0.00144	0.83
3	0.1	0.035	26.29	26.29	0.00	17.44	36.31	4.37	110.70	0.00133	0.37
4	11.9	0.045	73.98	0.00	2.63	0.54	53.72	4.62	135.49	0.00143	0.54

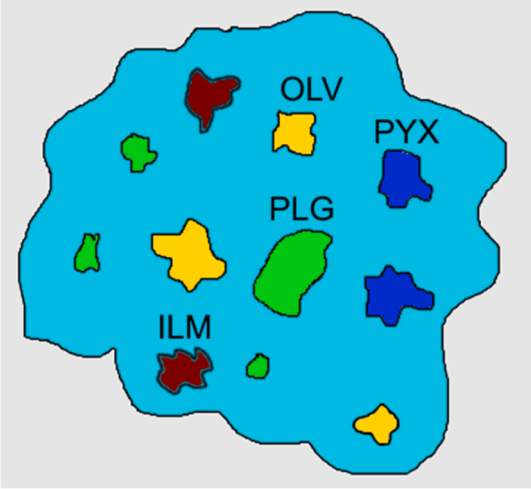

Fig. 11. Agglutinate grain with inclusions of basic minerals.

et al., 2001; Ohtake et al., 2013).

The results of the best fit using our algorithm are presented in Fig. 12 and Table 9. As one can see, our algorithm provides a good approximation for various types of spectra (relative error is in the range of 0.43 … 0.77%, mean value is 0.64%). The obtained concentrations of the basic minerals fall into the representative range for lunar materials, i.e., show geologically plausible values. The fitted parameter S is sensitive to the grain size of the lunar soil – at least a relatively young crater (site #1, Fig. 1) demonstrates a significantly higher value of the parameter S with respect to the rest of the points. The same is observed for the concentration of the reduced nanophase iron c_{nFe}: its value is lower for this young crater as compared to the undisturbed nearby mare area (site #2, Fig. 1). All the test points reveal a rather high concentration of plagioclase, especially for the highlands (points #3 and #4). Concentrations of clinopyroxene are found to be higher than those for orthopyroxene, which is also typical for lunar soils. Olivine is not detected besides the young crater (site #1). The predicted abundance of ilmenite is found to be moderate.

As can be seen from Table 9, the sum of the concentrations for the basic components Σc, including the gray component, is not exactly equal to 100%. This is due to the use of a fixed value for the albedo of the gray component. However, as was shown in Sec. 3.2, it practically does not affect the results of the unmixing procedure. Σc can easily be normalized to 100% by changing the albedo of the gray component, if that is necessary and/or important.

4.3. Shortcomings of the fitting algorithm

As pointed out earlier, numerical optimization procedures show a dependence of the results on the initial parameter values. This effect manifests itself not only for the parameter grain size in Table 10, where we present four different solutions for different initial values of c_{PLG} with approximately the same relative error for the test site #1 (see Figs. 2 and 9a). The same numerical behavior of the optimization procedure is observed for the rest of the test spectra. We explain this phenomenon by the impact of noise in spectra, especially when those do not have strong spectral features (as for the spectra shown in Fig. 9b–d). It is therefore important to use different initial values for all free parameters when working with different types of the lunar surface.

Using a priori information to regularize the fitting procedure can also be a possible solution to this problem. We have therefore decided to resolve this problem by following a different approach which makes use of the ANN technique (Sec. 5).

5. Mapping parameters of mineralogical composition of the lunar surface using the spectral mixing model

5.1. The idea of the mapping method

We have developed a new mapping method, which combines the nonlinear spectral mixing model of Shkuratov et al. (1999) with a correlation routine based on an artificial neural network (ANN). The idea of the method is the following. In the first step, we generate a set (pattern) of bidirectional reflectance spectra using our basic spectra set (Sec. 3) and spectral mixing model (Appendix) with random values of the mixture parameters. In a second step, we train the ANN with these spectra. And finally in a third step, we apply the trained ANN to the Chandrayaan-1 M^3 hyper-spectral data to map the compositional and structural parameters of the lunar surface.

5.2. Generation of a spectra set (pattern) for the ANN training

As outlined above, a training set containing 20,000 (the value is selected empirically) spectra are generated with random values of 8 mixture parameters: 5 × $c(j)$, c_{Gr}, c_{nFe}, and S. On the first iteration, a uniform distribution of random values is used. The initial ranges of variation for concentrations and grain size are chosen with a small margin (20%) from the values obtained for several (10–20) characteristic points of the studied area using the numerical fit procedure described above. Various training sets can be generated by varying the 8 mixture parameters through a change of the allowed ranges, to select the best variant, which provides an adequate mapping of the surface parameters. For different areas of the lunar surface, these ranges may be different depending on the compositional and morphological properties of these areas. Empirically selected ranges for the area shown in Fig. 3 are presented in Table 11. Note that the optimal selection of the ranges for the parameters for the generation of the training set provides some regularization of the prediction procedure. Solutions are searched within the defined "corridor" of the compositional parameters. For instance, for the test area, the variations of orthopyroxene and olivine concentrations are restricted to a narrow range near zero, that is characteristic for the typical lunar surface.

We assume that it is possible to construct a spectral training set which is suitable for the entire lunar surface. But such a universal set would be too big, not computationally effective, and would probably give inaccurate solutions for specific areas of the lunar surface.

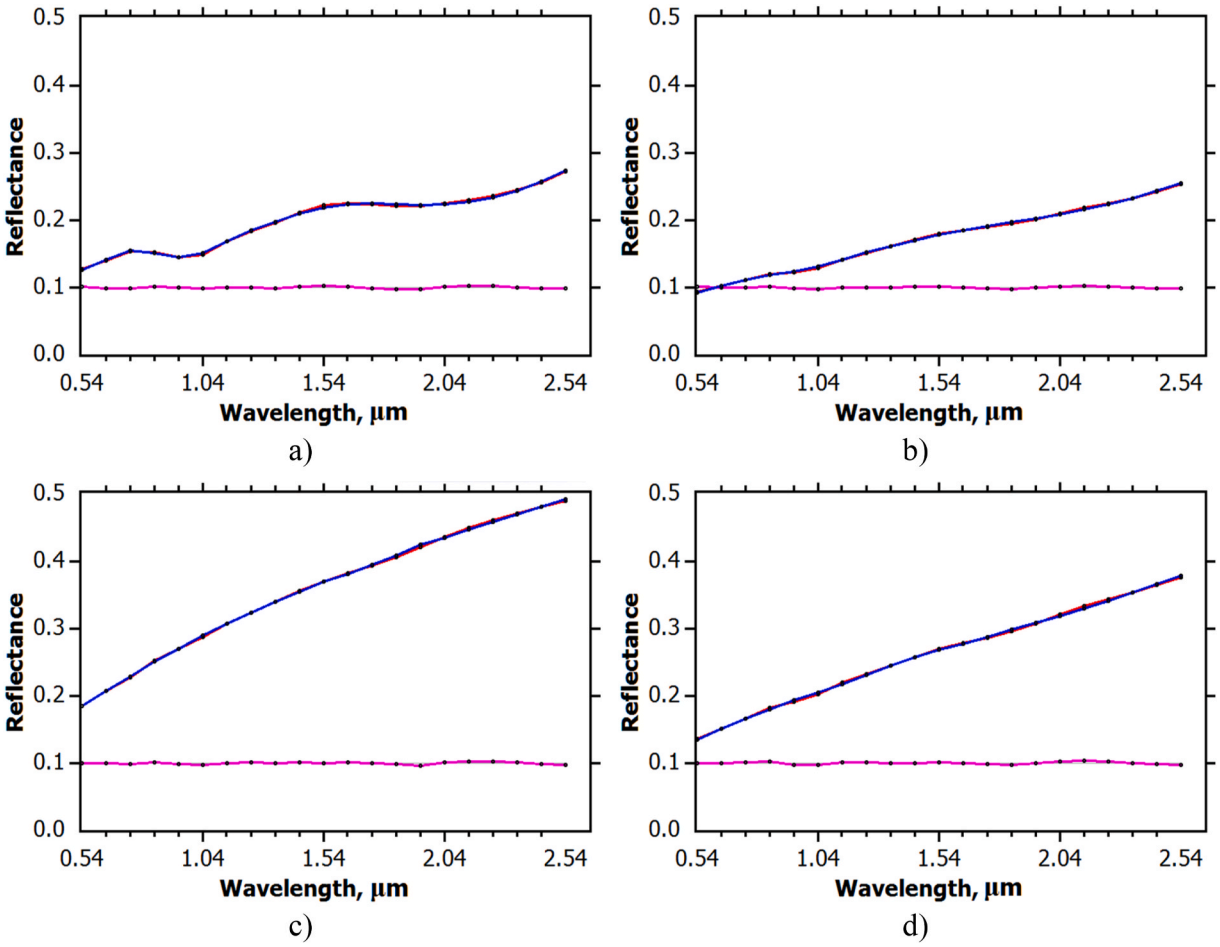

Fig. 12. Application of the spectral mixing model to Chandrayaan-1 M^3 spectra: (a) point #1; (b) point #2; (c) point #3; d) point #4. Blue is a measured spectrum; red is a calculated spectrum; magenta is a difference between calculated and measured spectrum +0.1. (See also Table 7).

Table 9
The results of applying the fitting procedure to Chandrayaan-1 M^3 spectra. $<A>$ is the mean albedo (see Fig. 3), σ is the standard error and $\sigma_{Rel} = 100\%\ \sigma/<A>$ is the relative error of the spectrum approximation.

Site	S, μm	c_{nFe}	c_{Gr}, %	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	Σc, %	σ	σ_{Rel}, %
1	138.7	0.087	54.39	17.65	6.54	6.98	25.78	7.00	111.80	0.00155	0.77
2	88.4	0.139	43.53	18.88	0.01	0.00	30.61	13.24	106.27	0.00122	0.70
3	8.0	0.041	16.42	11.41	0.00	0.59	68.21	6.59	103.22	0.00154	0.43
4	38.1	0.071	37.89	11.53	1.48	0.00	48.40	11.57	110.87	0.00168	0.64

Table 10
The results of applying the fitting procedure to Chandrayaan-1 M^3 spectrum of test point #1 with different initial values of parameters of the numerical optimization procedure. $<A>$ is the mean albedo (see Fig. 3), σ is the standard error and $\sigma_{Rel} = 100\%\ \sigma/<A>$ is the relative error of the spectrum approximation.

Initial values of c_{PLG},%	S, μm	c_{nFe}	c_{Gr}, %	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	σ	σ_{Rel}, %
0	158.7	0.083	50.69	12.93	9.91	6.62	32.37	4.63	0.00161	0.80
10	137.4	0.086	54.35	17.71	6.48	7.00	25.60	7.06	0.00155	0.77
20	138.7	0.087	54.39	17.65	6.54	6.98	25.78	7.00	0.00155	0.77
50	116.9	0.084	54.74	19.00	5.58	7.52	21.93	7.88	0.00158	0.79

5.3. ANN: training and mapping procedures

Following our previous work (Korokhin et al., 2008) we use here a kind of artificial neural networks – multilayer perceptron (Rumelhart and McClelland, 1986) with an error back-propagation-learning algorithm (Haykin, 1994). Empirically, we have found that the best result is obtained using an ANN with two hidden layers (Fig. 13) with the following parameters.

1) the number of input neurons is 21, corresponding to the number of points in an input spectrum (original M^3 spectra are resampled to a wavelength step $\Delta\lambda = 0.1$ μm to reduce the size of the

Table 11
Parameters for generation of a spectra set (pattern) for training the ANN. N = 20,000.

Range	S, μm	c_{nFe}	c_{Gr}, %	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %
Min	0	0.0	20	0	−2	−2	0	0
Max	120	0.2	40	50	5	5	35	20

processed data. Various tests have shown that this does not affect the unmixing results, but reduces the computation time);
2) the number of output neurons is 8, corresponding to the number of free parameters of our spectral mixing model (5 × $c(j)$, c_{Gr}, c_{nFe}, and S);
3) the number of hidden layers is 2;
4) the number of neurons in hidden layers are selected empirically;
5) the hidden layers contain a bias-neuron;
6) the sigmoid activation function $f(y) = 1/(1 + \exp(-\alpha y))$ with $\alpha = 1.0$ is used;
7) the input spectra are normalized into the [-1,+1] range. During testing, we use two kinds of normalizing procedures: (1) data are normalized for each wavelength independently, at the same time a certain compensation of the general slope of lunar spectra is performed; (2) all spectra in the data set are normalized with shared coefficients. Tests showed that the first method gives better results. Normalization coefficients are calculated from the training set (see Sec. 5.2);
8) output data (5 × $c(j)$, c_{Gr}, c_{nFe}, and S) are normalized to a 0.05 … 0.95 range using factors calculated from the pattern mixture parameters array (see Sec. 5.2);
9) an adaptive or constant training rate is used. In the adaptive case, the rate is calculated as the product of a relative prediction error in the previous training epoch by some factor k (typical values are from 0.01 to 1.0, selected empirically);
10) the inertia moment (Haykin, 1994) for the training is 0.9.

We need to discuss the question which criterion one has to implement to determine that the training process has been successfully terminated? An undertrained ANN will not take into account all the details in the correlation relations between input and output data (see Fig. 14a and d). An overtrained ANN will be sensitive both to real relations and to noise (see Fig. 14c and f). As a rule, a fixed number of training epochs is used (for instance, as in our work (Korokhin et al., 2008)) or the training is stopped when the rate of decreasing the error of the solution on the training pattern becomes less than the defined threshold value. In the present work, we propose to use an alternative approach. We consider the training process as successful when the ANN provides the optimal fit on average for the M^3 spectra for all pixels which were used in the training set (see Figs. 15b and 14b, e). The decision to stop the algorithm's learning is taken by the researcher interactively based on the analysis of the behavior of the dependence of the average relative error of the approximation of the M^3 spectra on the number of the training epoch (see "Optimum" in Fig. 15b).

Parameters of the ANN which provide the best results for mapping the mineral composition of the lunar surface are presented in Table 12. As we can see from the table, the time needed for training and mapping is significantly shorter compared to using a numerical optimization algorithm (see Sec. 2.3, 5.1). The typical training time to obtain the optimal solution is in the range of 1–60 min on an AMD Ryzen 9 3950X processor, dependent on the configuration of the ANN and the size of the training set. The mapping is very fast (less than 1 s for a typical M^3 image). The dramatic increase in the computational speed allows us to check a significant number of different ANN configurations and training sets, in order to choose the optimal one. It should be noted that a repetition of the global minimum search with the same surface region and the same computational resources using the standard numerical optimization procedures requires orders of magnitude more time (about 35 h).

5.4. Testing the ANN algorithm with selected Chandrayaan-1 M^3 spectra

While testing our algorithm, we have checked several tens of different ANN configurations and training sets. As a result, we have chosen the best variant presented in Tables 11 and 12. In Table 13, the results from applying the ANN algorithm to the M^3 spectra of the different test sites (Fig. 3) are given. The ANN approach provides a fairly good approximation for different types of spectra (Fig. 16), excluding sites #1 and #3, corresponding to the young crater and the high albedo highland area with a featureless spectrum, correspondingly. The formal fitting quality is in the particular case lower for the ANN method than it is for a numerical optimization (Figs. 16 and 12). However, the ANN algorithm is closer to a global minimum in the modeled parameter space for the whole set of spectra. Note, averaged over the whole test area (Fig. 14e) the relative error is 2.393% (Table 12).

The obtained concentrations of the basic minerals show realistic values. The calculated parameter S is sensitive to the grain size of the lunar soil. A relatively young crater (point #1) has a significantly higher value of the parameter S relative to the rest of the points. The concentration of the reduced nanophase iron c_{nFe} has a lower value here, as we expect for young formations.

Compositional parameters calculated using the ANN (Table 13) differ to a certain extent from the obtained parameters obtained by numerical optimization (see Table 9), although the difference between the concentrations is rather small. The biggest difference is observed for the site #1 (young crater). Most likely the problem arises from the

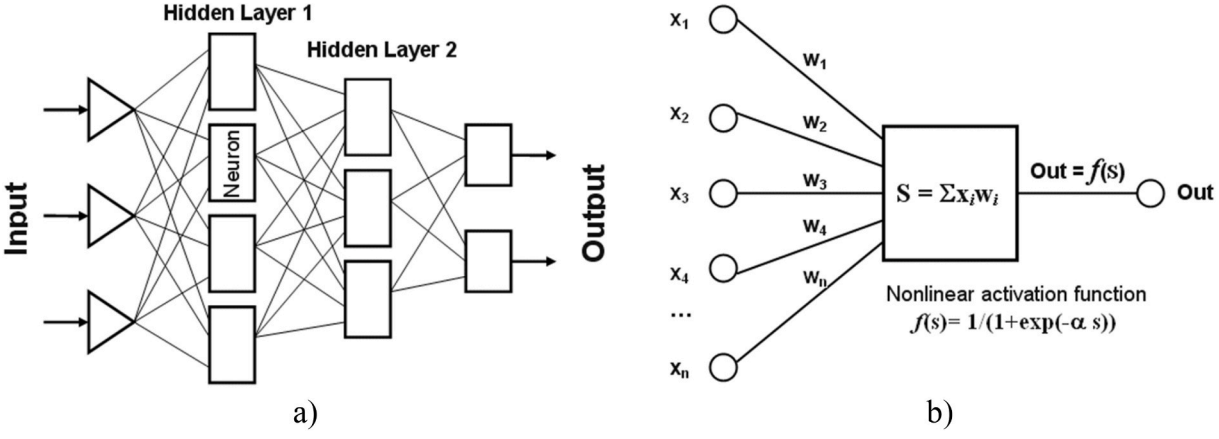

Fig. 13. (a) Multilayer perceptron used for mapping; (b) scheme of an artificial neuron.

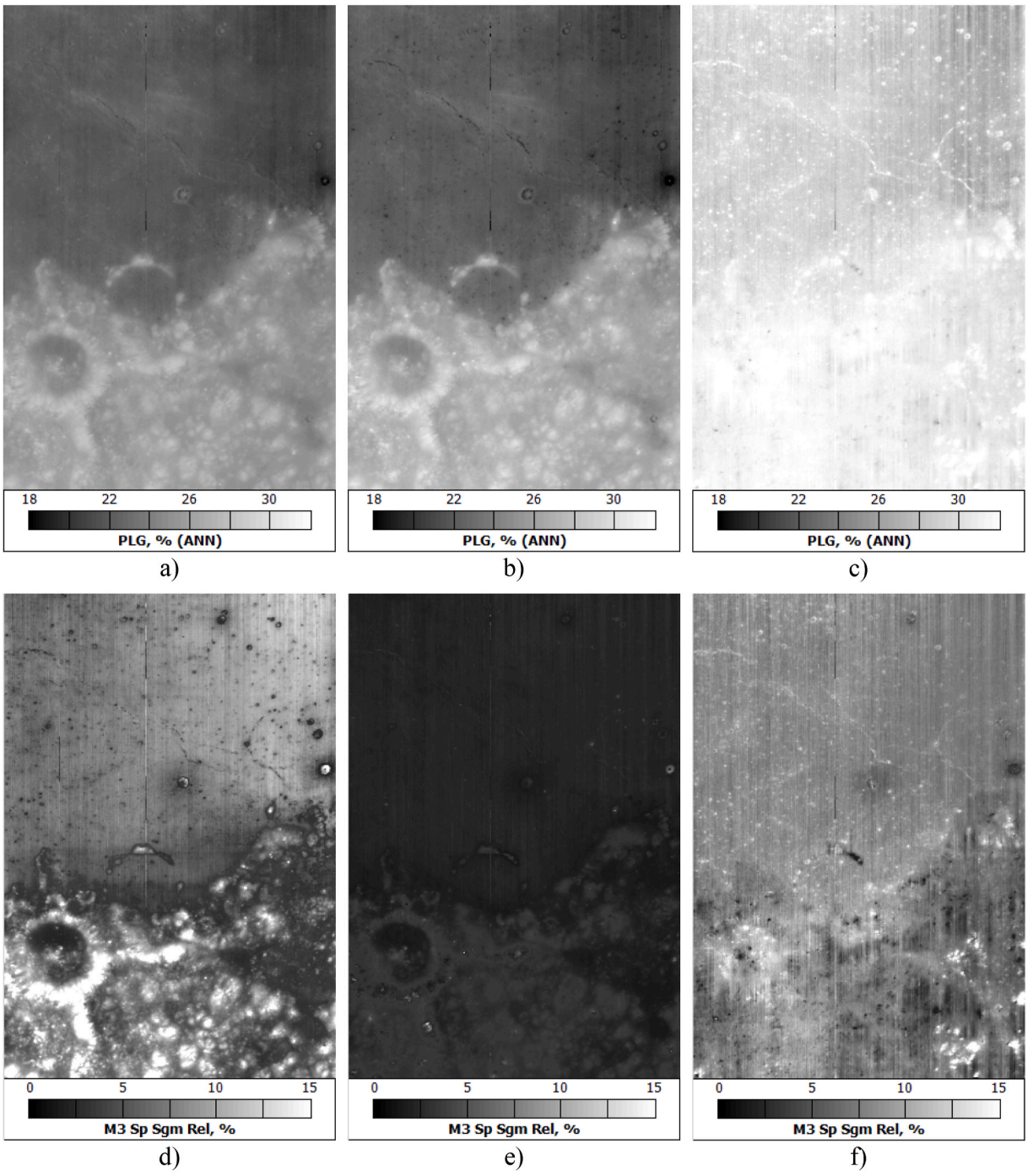

Fig. 14. Plagioclase concentration (c_{PLG}) maps and relative error of spectra approximation (σ_{Rel}) constructed using undertrained ANN (10 epochs) (a, d), optimally trained ANN (66 epochs) (b, e) and overtrained ANN (200 epochs) (c, f) (iteration #1, Table 11).

ambiguity of the numerical optimization solution for data which contain a significant amount of noise (see Sec. 4.3). Note that fixing the values of one or two parameters (for instance, S and c_{PLG}) allows the numerical fit procedure to obtain a solution closer to the one obtained by the ANN. This confirms our assumption.

5.5. Iterative procedure of the ANN training

Fig. 17a and d presents the results of mapping the test area (Fig. 3) with the ANN technique. As one can see from Fig. 17d, the method provides good results throughout the whole area. The average relative error of the approximation is 2.39%. Errors are greater for some young craters and highland areas with have large relief slopes (up to 5%). Highland areas show higher error values compared to mare areas, on average, 3% versus 2.2%. This is probably caused by deficiencies related to the choice of the spectra in the training set. Responsible for this is the use of a uniform random distribution of the mixture parameters which was needed for the generation of the training set (see Sec. 5.2), while in reality they are distributed in a complex way. This leads to the appearance of redundant information in the training set.

To avoid this problem, we performed additional training runs. We generated a new training set after each step using the results from the previous iteration (but added only small random disturbances, less than ±0.5%) instead of generating new spectra from new random mixture

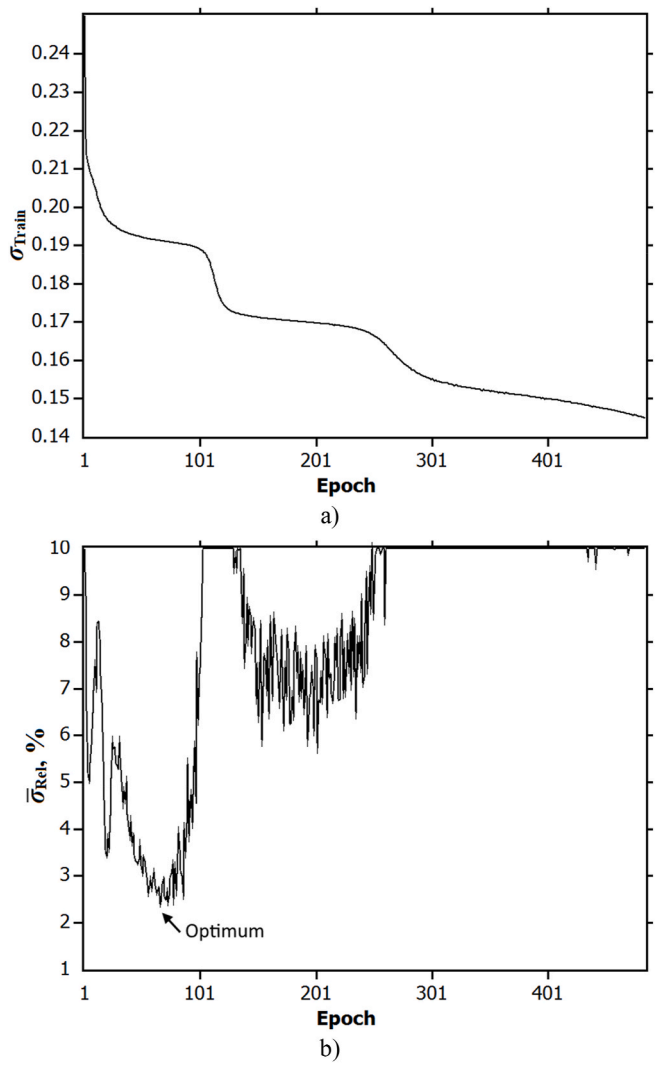

Fig. 15. Average normalized error on the training pattern (a), and average relative error of the M^3 spectra approximation (b) as a function of the training epoch (Table 11, Iteration #1).

parameters. A random mixture is used only for the first iteration. The training set has 152,000 entries, corresponding to the number of points in the M^3 image. The configuration of the ANN for the second iteration and parameters of the training are given in Table 12. The results of applying such an approach to the test area is presented in Table 14. As we can see, errors of the spectral approximations are significantly reduced, especially for the "marginal" sites #1 and #3. The mean relative error for whole scene is 1.26%, in contrast with 2.39% for the first iteration. In the map of the relative error (Fig. 17e), only young craters and some extended formations (σ_{Rel} up to 2.5–3.0%) remain visible. Other morphological features do not significantly appear on this map. Errors for the highland areas turn out to be lower in comparison with mare areas, on average 1.0% versus 1.5%. This indicates that such a training set is better suited for training the ANN than the initial random set.

To lower the relative error again iterations can be repeated. Our experiments showed that for test area, 3 iterations are enough to achieve good results (see Tables 12–15, Fig. 17f and 18). A fourth iteration does not give significant better results.

As we can see in Fig. 17f, the third iteration gives a uniformly good approximation of the observed spectra over the whole scene ($\boldsymbol{\sigma_{Rel}} = 1.12\%$). Only some young craters stand out in the background. Highland areas display lower relative errors due to they are calculated as the division of the standard error by the mean albedo (see Fig. 3).

A comparison of the results obtained by the ANN algorithm (Table 15) with the results of the fitting procedure (Table 9) shows that, in general, proportions of the concentrations of the basic components are similar. But in details the results are different. The biggest difference is observed for the parameter *S*, which can lead to a difference in the concentrations of the basic elements and the "gray component". Such a difference may be caused by applying the fitting algorithm to lunar spectra that do not have strong spectral features, and this results in the algorithm finding the wrong local minima of the functional (see discussion in section 4.3). Note that the fitting procedure better approximates the spectra (cf. Values of σ_{Rel}), which does not necessarily mean that the algorithm has found the true minimum.

5.6. *Discussion of mapping results using the ANN technique*

In Fig. 19a and b, maps of grain size *S* and concentration of nFe^0 are shown. They show a realistic parameter distribution. Mare areas (upper part of the scene) exhibit an inverse correlation between *S* and c_{nFe}. In particular, young craters show tangibly increased values of *S* and decreased values of c_{nFe}. The ejecta area around the mare crater near the

Table 12
Parameters of ANN used in this study for mapping the mineral composition of the lunar surface. *N* [1] and *N* [2] denote the number of neurons in the first and second hidden layer respectively. Adaptive training rate *k* is a coefficient, used for the adaptive calculation of the training rate. σ_{Rel} is average relative error of spectra approximation. t_{Epoch} is the time for one epoch of training, t_{Optim} is the time for finding the optimum in the training run, t_{Map} is the time used for mapping, using a trained ANN (all times are given for a AMD Ryzen 9 3950*X* processor with 128 GB memory).

Iteration	*N* [1]	*N* [2]	Initial training rate	Adaptive training rate *k*	Pattern size	σ_{Rel}, %	Optimal epoch	t_{Epoch}, sec	t_{Optim}, min	t_{Map}, sec
#1	42	12	1.000	0.01	20,000	2.39	66	0.24	0.25	0.2
#2	42	12	0.002	–	152,000	1.26	5336	0.88	78.26	0.2
#3	42	12	0.010	–	152,000	1.12	4256	0.88	62.42	0.2

Table 13
Results of application of the ANN algorithm (Iteration #1) to Chandrayaan-1 M^3 spectra. $<A>$ is the mean albedo (see Fig. 3), σ is the standard error and $\sigma_{Rel} = 100\% \sigma/<A>$ is the relative error of the spectrum approximation. Mean relative error $\boldsymbol{\sigma_{Rel}} = 2.39\%$.

Site	*S*, μm	c_{nFe}	c_{Gr}, %	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	Σc, %	σ	σ_{Rel}, %
1	67.9	0.041	31.19	33.02	1.65	1.64	19.18	15.22	100.25	0.00812	4.05
2	45.6	0.103	32.31	25.46	1.43	1.55	20.96	16.29	98.00	0.00368	2.13
3	4.0	0.042	31.78	36.90	1.39	1.24	30.18	8.46	109.95	0.01315	3.67
4	17.7	0.055	31.76	32.41	1.46	1.38	26.78	12.98	106.97	0.00704	2.68

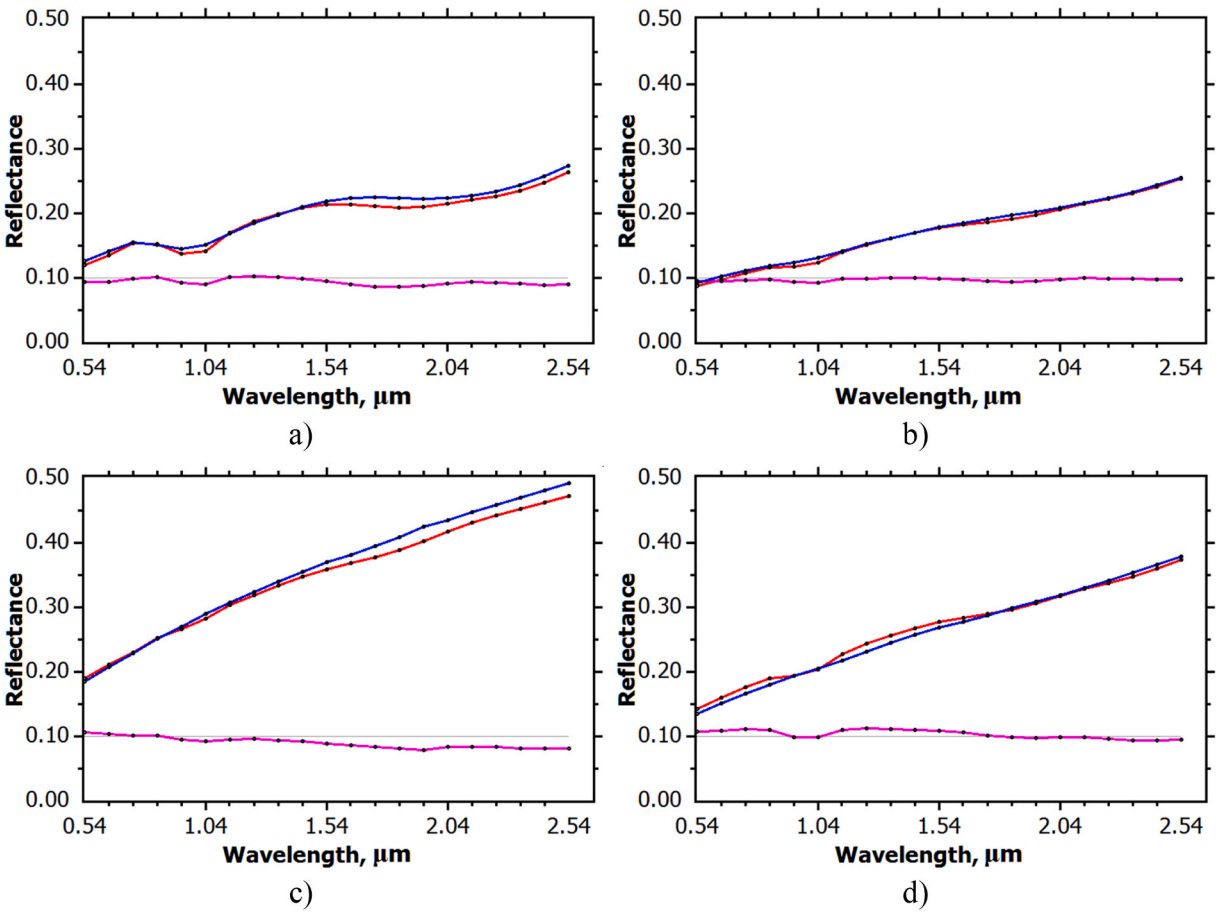

Fig. 16. Application of the ANN algorithm (Table 11, Iteration #1) to Chandrayaan-1 M^3 spectra: (a) point #1; (b) point #2; (c) point #3; (d) point #4. Blue line shows a measured spectrum; red line displays a calculated spectrum; magenta line shows the difference between the calculated and the measured spectrum +0.1.

right edge of the frame also exhibits slightly higher S and lower c_{nFe}. These are consistent with the results obtained by the phase-ratio method (see, for instance, Kaydash et al., 2012, 2014, 2018; Shkuratov et al., 2011).

In contrast to this, highland regions (lower part of the scene) show a direct correlation. This can be explained as follows: (1) for the highlands, space weathering processes reach saturation; (2) properties of the highlands regolith are different compared to the regolith located in mare regions, which leads to a different formation intensity of reduced iron; (3) in Fig. 19b, we can observe a low concentration of nFe^0 in areas with contain possible rock outcrops, where the depth of the regolith layer is low or zero; (4) the Shkuratov et al. spectral mixing model is not appropriate for highlands soil, and especially for rocky areas.

A quantitative comparison of the grain size and maturity maps which was obtained by an alternative method is discussed below.

The most abundant basic mineralogical components are clinopyroxene (Figs. 19d and 20d), plagioclase (Figs. 19g and 20g) and ilmenite (Figs. 19h and 20h). The content of orthopyroxene (Fig. 19e and 20e) and olivine (Fig. 19f and 20f) is very low; for most of the surface it is almost zero. This is consistent with the geological type of the test area and the results of the preliminary analysis using the numerical optimization approach (see Sec. 4.2).

In Fig. 21, the correlation diagrams "Concentration of basic component vs. $<A>$" are shown (A is albedo averaged in the spectral range of 0.54 … 2.6 μm and characterizes the general reflectance properties of the lunar surface). Such diagrams can be used as a tool for clustering the data into separate mare and highland regions. All diagrams have two branches formed by mare and highland areas, that corresponds to a bimodal distribution of all resulting parameters, excluding the orthopyroxene (see Fig. 20). The branches are well distinguished, excluding "PLG vs. $<A>$", for which the highland branch extends the mare one. At the same time, the correlation is much higher for the highland areas. A significant linear correlation (0.98) is observed for clinopyroxene (Fig. 21b). Plagioclase abundance is also highly correlated with albedo, but the nature of this correlation is nonlinear for the highland branch. Olivine and ilmenite show some inverse correlations with albedo for the highland, while for mare regions, the correlation is absent. Considering the large abundance of CPX, PLG and ILM for the test scene, it can be concluded that the distribution of albedo is controlled mainly by these minerals, at least for the highland regions. We also note that the plots in Figs. 21b and f may be affecting each other because clinopyroxene and ilmenite spectra can both make a red slope at the 2-μm region. The gray component also shows some correlation with albedo (Fig. 21a). But c_{Gr} varies in a very narrow range (see Fig. 20e), and can be considered as constant.

5.7. Comparison with the results of other methods and discussion

We have compared our results with other methods using spectrophotometric data for a determination of the mineralogical composition, the grain size distribution and the maturity of the lunar regolith. The first method is based on studying the direct statistical relationships between spectral characteristics and the chemical/mineral composition of lunar soils (Shkuratov et al., 2003a, 2003b, 2005a, 2005b, 2007; Pieters et al., 2006). Data from the Lunar Soil Characterization Consortium (LSCC) were used (Pieters et al., 2000, 2002, 2006; Taylor et al., 2001a, 2001b). The method assumes that the mineralogical/chemical composition, maturity and grain size of the lunar soils linearly correlates with

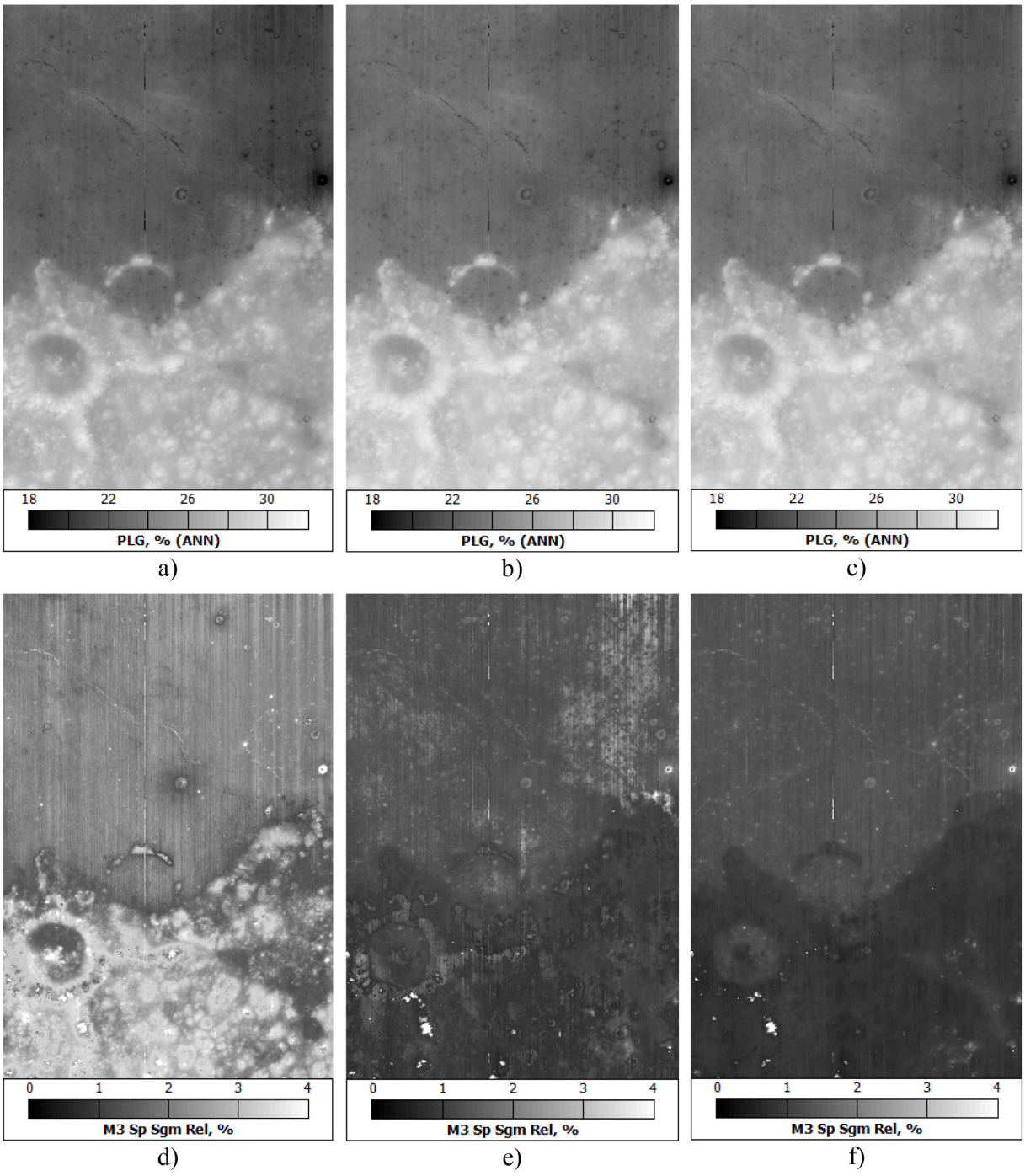

Fig. 17. Plagioclase concentration (c_{PLG}) maps of and relative error of spectra approximation ($\boldsymbol{\sigma_{Rel}}$) for 3 consecutive iterations (See Table 11): (a, d) iteration #1; (b, e) iteration #2; (c, f) iteration #3.

Table 14
The results of applying the ANN algorithm (Iteration #2) to Chandrayaan-1 M^3 spectra. $<A>$ is the mean albedo (see Fig. 3), σ is the standard error and $\sigma_{Rel} = 100\% \sigma/<A>$ is the relative error of the spectrum approximation. Mean relative error $\boldsymbol{\sigma_{Rel}} = 1.26\%$.

Site	S, μm	c_{nFe}	c_{Gr}, %	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	Σc, %	σ	σ_{Rel}, %
1	57.0	0.043	31.31	32.30	1.60	1.59	20.51	13.54	99.25	0.00439	2.19
2	35.1	0.096	32.30	23.86	1.38	1.48	22.15	15.78	96.95	0.00377	2.18
3	1.0	0.048	31.86	37.66	1.37	1.21	30.70	7.33	110.13	0.00496	1.38
4	9.7	0.063	31.99	29.72	1.39	1.31	28.23	13.98	106.62	0.00243	0.92

Table 15
The results of applying the ANN algorithm (Iteration #3) to Chandrayaan-1 M^3 spectra. <A> is the mean albedo (see Fig. 3), σ is the standard error and $\sigma_{Rel} = 100\% \sigma/<A>$ is the relative error of the spectrum approximation. Mean relative error $\boldsymbol{\sigma_{Rel}} = 1.12\%$.

Site	S, μm	c_{nFe}	c_{Gl}, %	c_{CPX}, %	c_{OPX}, %	c_{OLV}, %	c_{PLG}, %	c_{ILM}, %	Σc, %	σ	σ_{Rel}, %
1	55.4	0.041	31.29	31.710	1.60	1.58	20.74	13.76	99.08	0.00434	2.17
2	35.6	0.095	32.31	23.29	1.37	1.48	21.75	15.97	96.17	0.00202	1.17
3	1.4	0.046	31.78	38.29	1.39	1.22	30.63	6.82	110.13	0.00224	0.63
4	10.6	0.063	31.97	29.85	1.40	1.32	28.07	13.46	106.07	0.00241	0.92

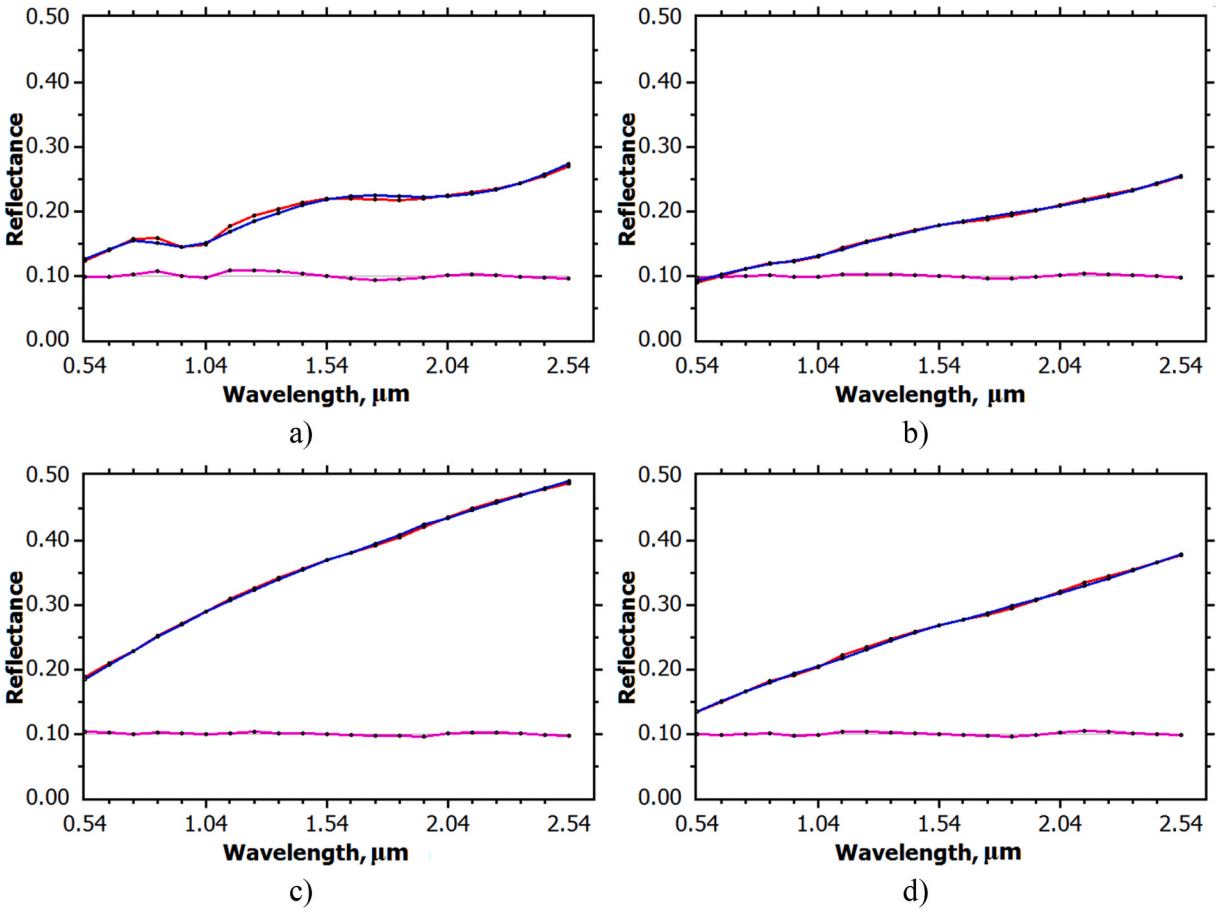

Fig. 18. Application of the ANN algorithm (Table 11, Iteration #3) to Chandrayaan-1 M^3 spectra: (a) point #1; (b) point #2; (c) point #3; (d) point #4. Blue line shows a measured spectrum; red line shows a calculated spectrum; magenta line displays the difference between calculated and measured spectrum +0.1.

the spectral parameters corresponding to photometric bands of Clementine CSR UVVIS camera (Nozette and the Clementine Team, 1994). Such a relation can be described by a multiple linear regression (MLR) between the studied parameter P (concentration, maturity or grain size) and reflectance at several spectral channels as, for instance, proposed in Shkuratov et al. (2005a); Pieters et al. (2006):

$$\log P = a_0 + a_1 A(415) + a_2 A(750) + a_3 A(900) + a_4 A(1000), \quad (3)$$

where a_i are the weight coefficients, $A(\lambda)$ are reflectances in % corresponding to the Clementine CSR UVVIS photometric bands (Nozette and the Clementine Team, 1994), λ is in nanometers.

Instead of reflectances, it is possible to use derived spectral parameters as, for example, proposed in Shkuratov et al. (2003a); Pieters et al. (2006):

$$\log P = b_0 + b_1 A_R + b_2 C_{IR1} + b_3 C_{IR2} + b_4 D, \quad (4)$$

or in Shkuratov et al. (2003b):

$$\log P = c_0 + c_1 A_R + c_2 C_{BR} + c_3 C_{IR1} + c_4 C_{IR2} + c_5 C_{IR3} + c_6 D, \quad (5)$$

where b_i and c_i are the weight coefficients, A_R, C_{BR}, C_{IR1}, C_{IR2}, C_{IR3}, D are spectral parameters derived from initial Clementine CSR UVVIS data reflectances: reflectance $A_R = A(750)$, color indices: $C_{BR} = A(415)/A(750)$, $C_{IR1} = A(900)/A(750)$, $C_{IR2} = A(950)/A(750)$, $C_{IR3} = A(1000)/A(750)$, and bend $D = A(750)A(950)/[A(900)]^2$. The logarithm in Eqs. (3)–(5) is used to avoid negative values of the predicted parameter. The weight coefficients are found using the minimal least-squares method for the LSCC data (Pieters et al., 2000, 2002, 2006; Taylor et al., 2001a, 2001b).

In Shkuratov et al. (2005a) and Pieters et al. (2006), several different variants of MLRs are considered and compared. We use Eq. (3) for mapping the maturity degree I_S that is the amount of nanophase reduced iron Fe0 in the regolith particles measured by ferromagnetic resonance techniques (e.g., Morris, 1976, 1978) (Fig. 22a), abundance of cliopyroxene (Fig. 23a), othopyroxene (Fig. 24a) and plagioclase (Fig. 26a). Eq. (5) is used for mapping the abundance of olivine (Fig. 25), and Eq. (4) for mapping the grain size (Fig. 27a), because these MLRs give better

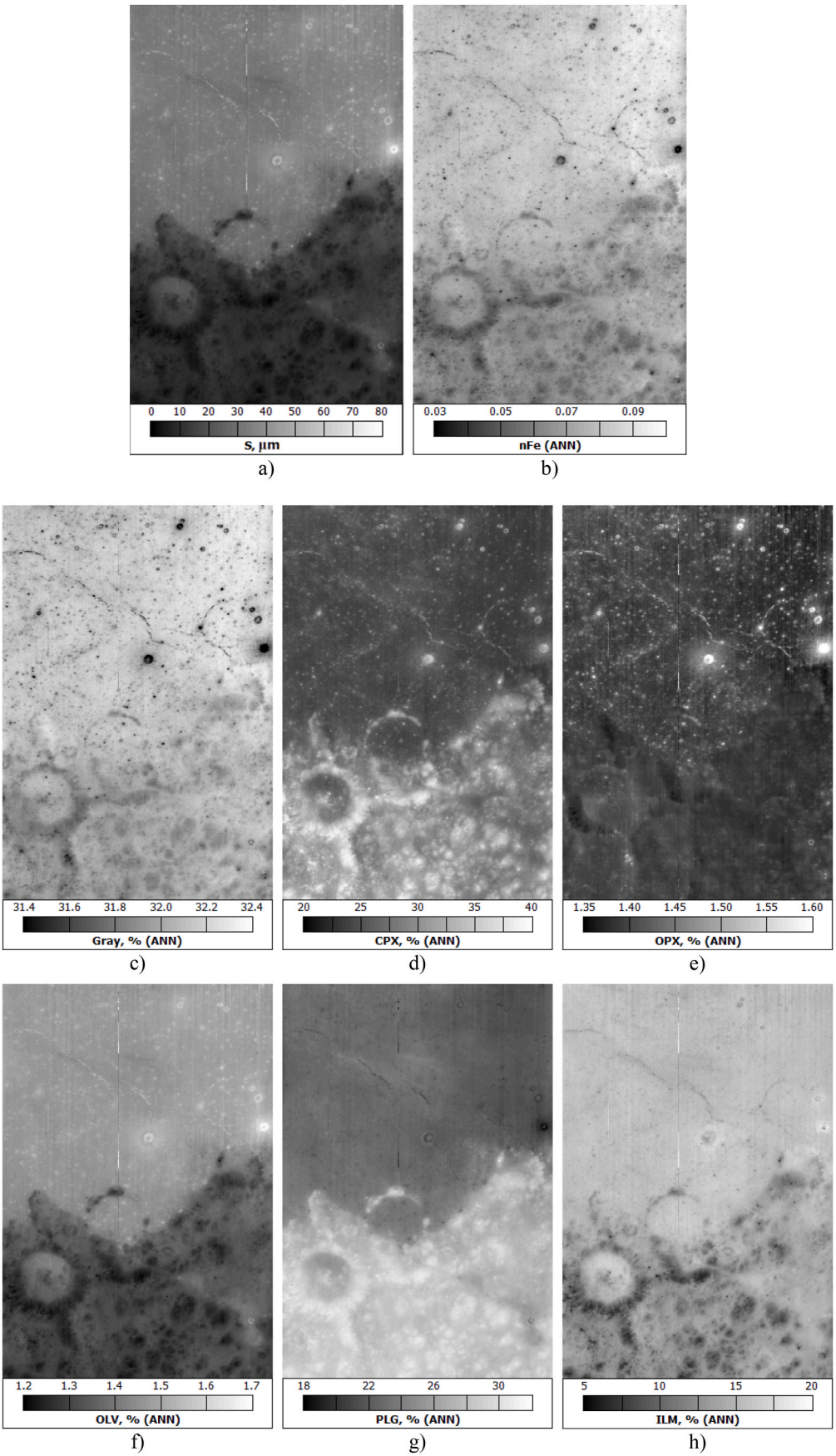

Fig. 19. Maps of grain size *S* (a), concentration of nFe^0 grains (b), gray component (c), and concentration of basic components (Table 3): (d) clinopyroxene; e) orthopyroxene; (f) olivine; (g) plagioclase; (h) ilmenite, obtained using the ANN mapping system (Table 11, Iteration #3).

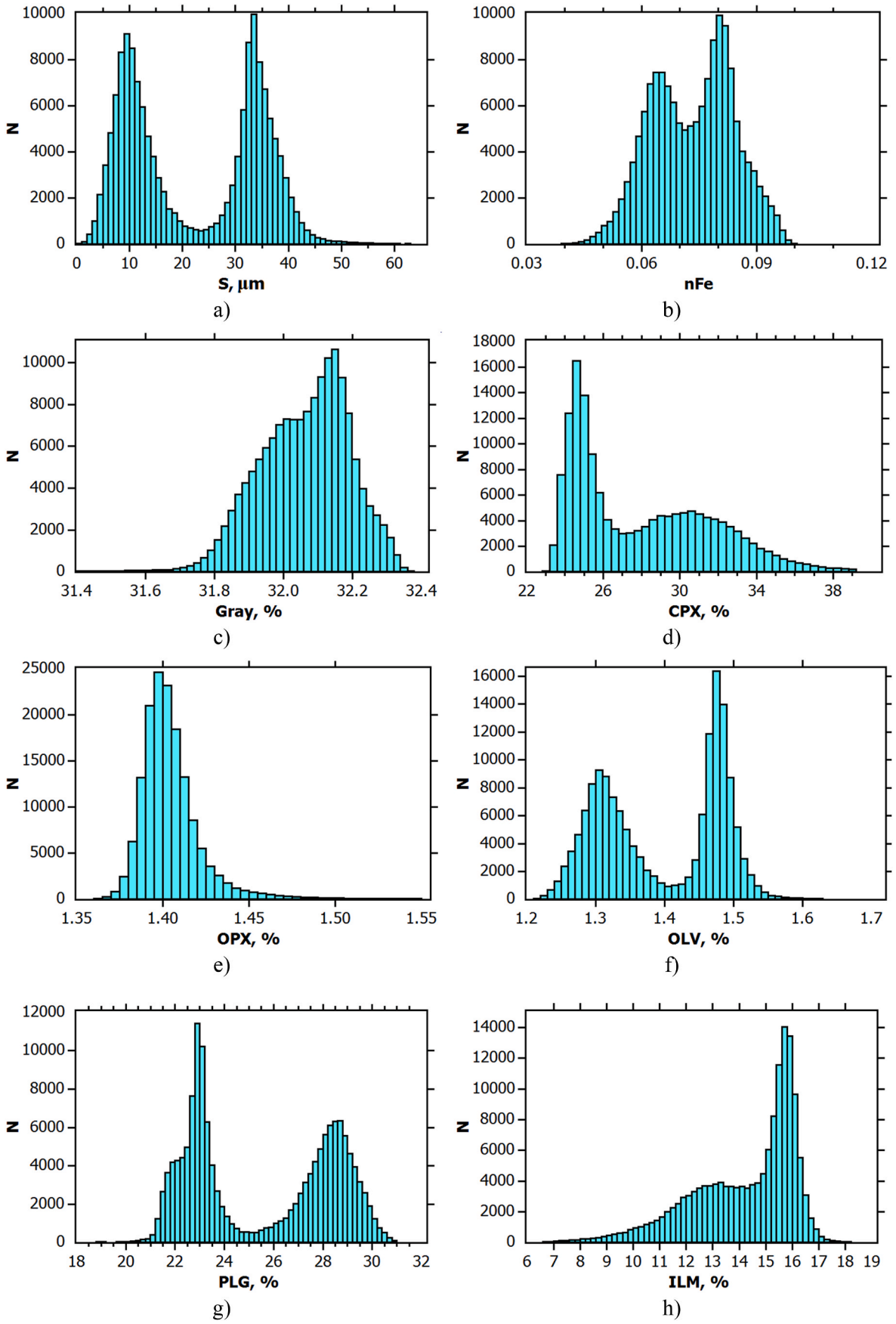

Fig. 20. Histograms of distribution of grain size *S* (a), concentration of nFe^0 grains (b), gray component (c), and concentration of basic components (Table 3): d) clinopyroxene; (e) orthopyroxene; (f) olivine; (g) plagioclase; (h) ilmenite, obtained using the ANN mapping system (Table 11, Iteration #3).

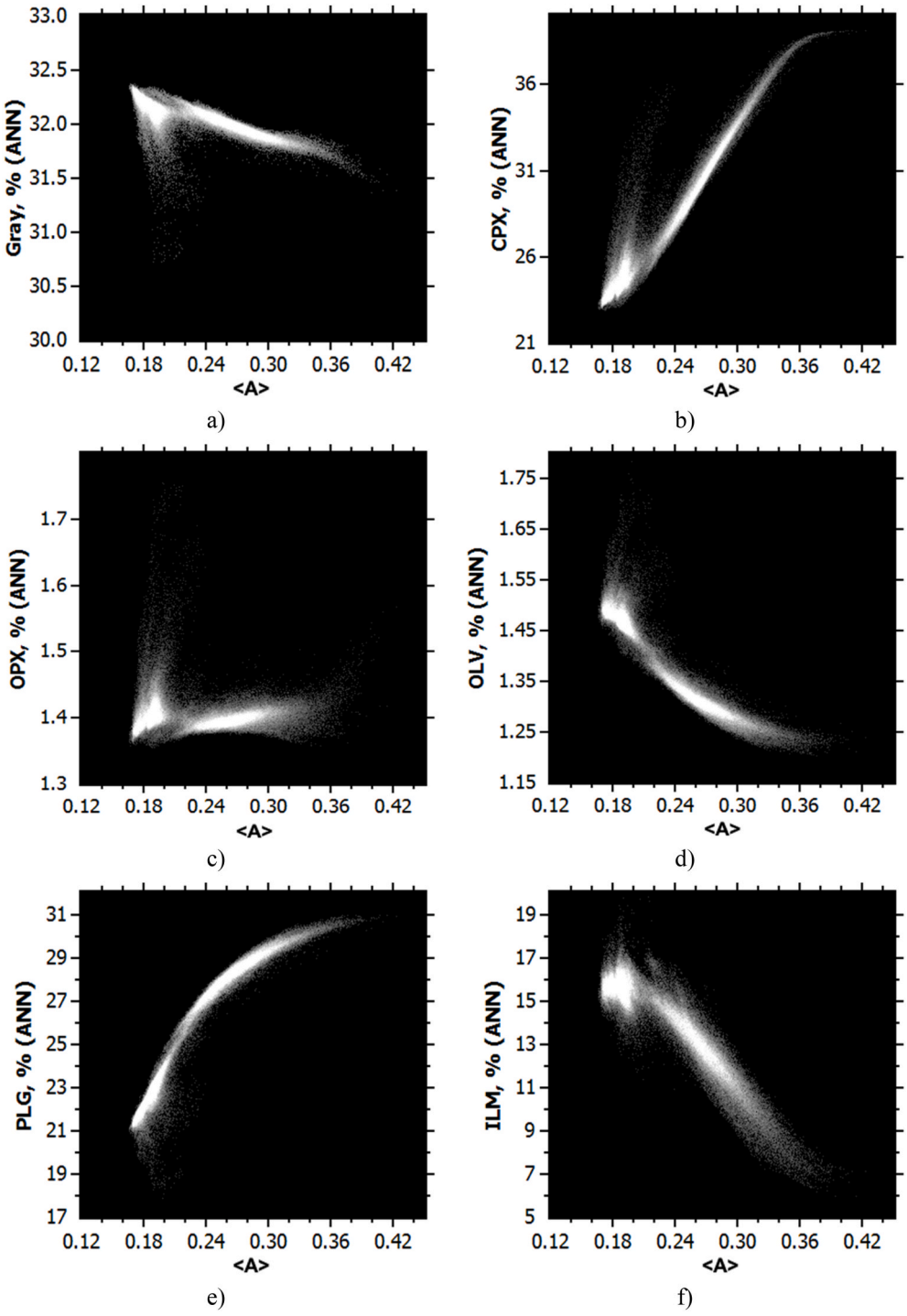

Fig. 21. Correlation diagrams "Concentration vs. mean albedo <*A*>" for: (a) gray component; (b) clinopyroxene; (c) orthopyroxene; (d) olivine; (e) plagioclase; (f) ilmenite.

results (maximum of correlation on LSCC data and minimum of noise and artifacts for the Chandrayaan-1 M^3 data). To apply these MLRs to M^3 data, we reduce M^3 spectra (converted into RELAB photometric system) to Clementine CSR UVVIS photometric bands *A*(415 nm), *A*(750 nm), *A* (900 nm), *A*(950 nm), *A*(1000 nm). To calculate *A*(415 nm) we extrapolated a linear regression relationship which was found by using the first 6 wavelengths channels (from 540.84 nm to 730.48 nm) of the M^3 spectra.

To be able to compare these results with the results from our new method, we have constructed correlation diagrams of the form "selected parameter (ANN) – selected parameter (MLR)".

As one can see, the correlation diagram "c_{nFe} (ANN) – I_S (MLR)" (Fig. 22b) shows a rather high correlation between the two maturity parameters ($c = 0.85$ for maria areas and $c = 0.83$ for highlands). Clearly visible is that the parameter regions for mare and highlands areas are separated and form parallel bands. Such a separation may indicate that for the determination of the mineralogical composition and maturity of the lunar soils from spectrophotometric data a linear approach may be insufficient.

There is also a very high correlation ($c = 0.99$) between the spectral

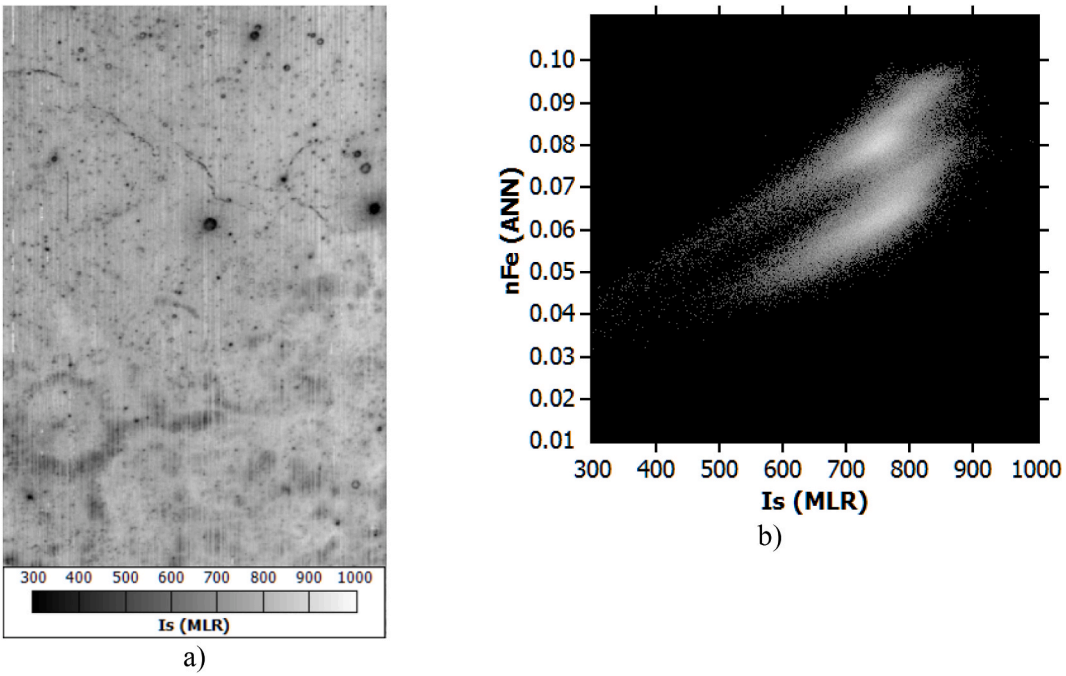

Fig. 22. (a) The degree of maturity (ferromagnetic resonance response) of the lunar soil I_S calculated using Multiple Linear Regression (MLR) obtained from LSCC data (Eq. (3)); (b) correlation diagram "c_{nFe} (ANN) – I_S (MLR)".

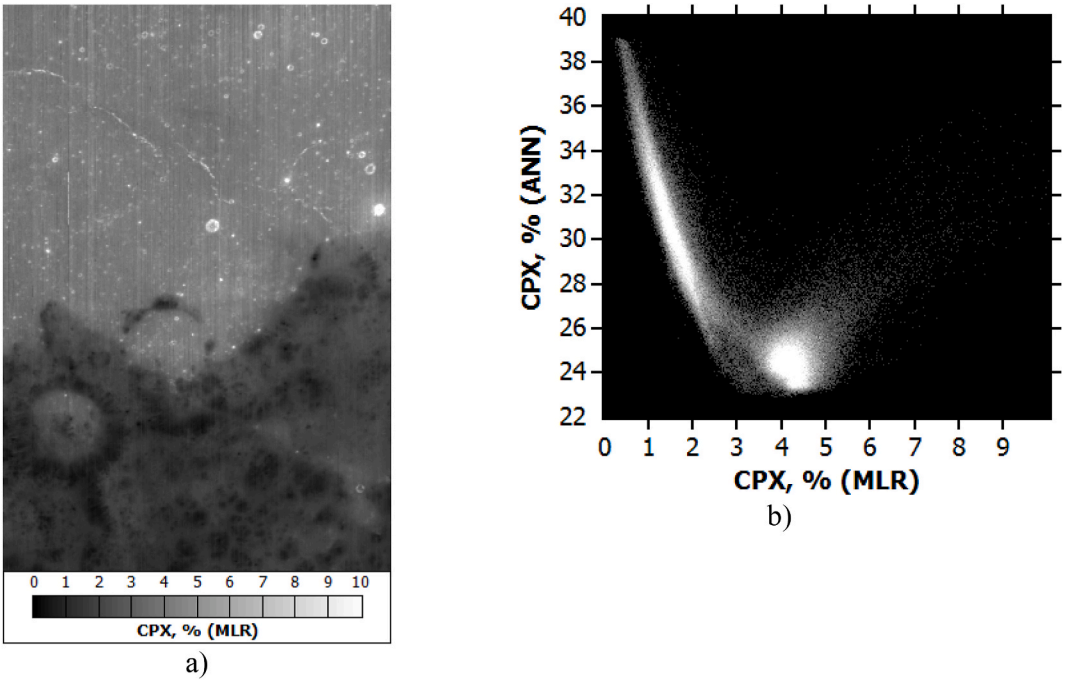

Fig. 23. (a) abundance of clinopyroxene calculated using Multiple Linear Regression (MLR) obtained from LSCC data (Eq. (3)); (b) correlation diagram "CPX, % (ANN) – CPX, % (MLR)".

mixing model parameter S (the average length of light propagation in a particle between internal reflections) and grain size d, obtained from MLR Eq. (2) (Fig. 27b). Moreover, mare and highland branches lie on the same regression line. And in general, the distribution of parameters is very similar on both maps (compare with Fig. 19a).

In Figs. 23–26, the results of a comparison of the ANN with the MLR-approach for clinopyroxene, orthopyroxene, olivine and plagioclase are shown. A significantly high correlation is observed only for plagioclase in the mare areas ($c = 0.92$) and for olivine in the highland areas ($c = 0.82$). The rest of the minerals either do not produce a direct correlation or even reveal an inverse correlation (clinopyroxene in the highlands). This can be explained by the fact that the assumption of the existence of a linear dependence of concentration of these components on spectral parameters for deriving equations (3)–(5) does not hold. Note, that in contrast to MLR, the ANN technique used in this work allows to find the nonlinear multiparameter regression (Rumelhart and McClelland, 1986; Korokhin et al., 2008). The MLR map of the plagioclase abundance is especially clear evidence for this – values of c_{PLG} for the highland areas

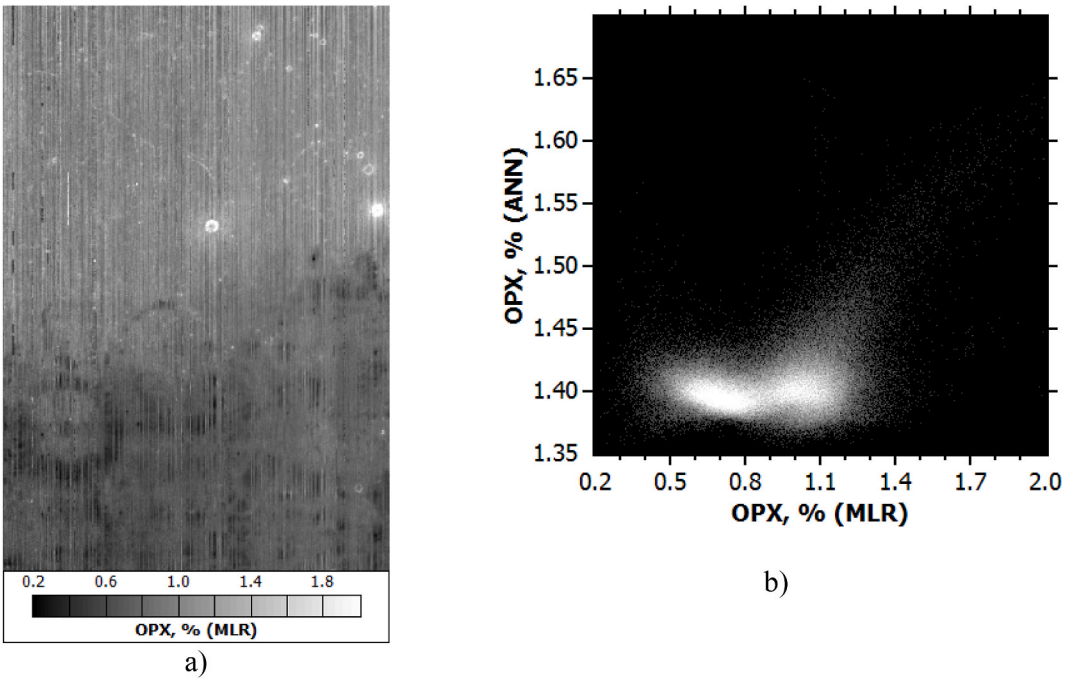

Fig. 24. (a) abundance of orthopyroxene calculated using Multiple Linear Regression (MLR) obtained from LSCC data (Eq. (3)); (b) correlation diagram "OPX, % (ANN) – OPX, % (MLR)".

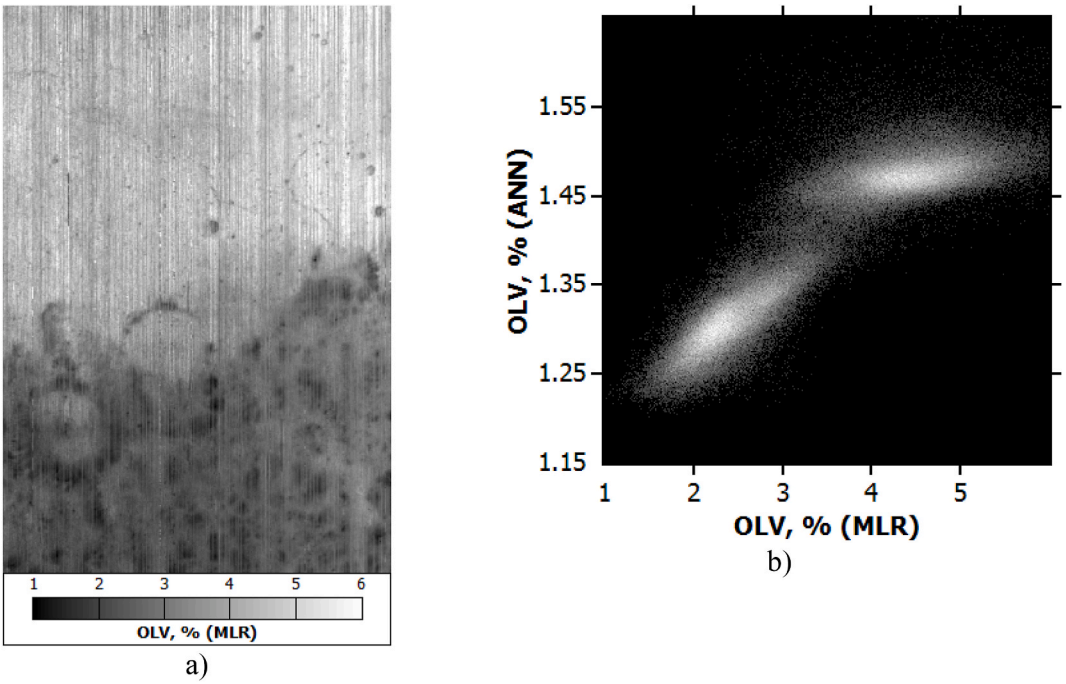

Fig. 25. (a) abundance of olivine calculated using Multiple Linear Regression (MLR) obtained from LSCC data (Eq. (5)); (b) correlation diagram "OLV, % (ANN) – OLV, % (MLR)".

are even higher than 100% (see Fig. 26a).

Let us point out that the mean values of the parameters are also different. In the case of clinopyroxene, olivine and plagioclase, a difference of 2–3 times is observed. Our approach gives smaller values. This can be explained partially by some uncertainties in the parameters of the spectral mixing model, such as the value of the real part of the refractive index n, the packing parameters q, and the grain size of the samples of the basic components. A direct comparison of spectrally obtained concentrations of the basic minerals with the values found in the laboratory is not entirely valid. The fact is that Taylor et al. (1996) measured the concentration of pyroxenes, olivine, plagioclase and ilmenite in separated grains of lunar LSCC-samples. Agglutinate particles were sorted separately. Measurements of their content showed very large values of mass concentrations (up to 60%, https://pgi.utk.edu/lunar-soil-characterization-consortium-lscc-data/). However, these agglutinate particles can include grains of pure basic minerals (Fig. 11). These inclusions can significantly modify the lunar spectra by changing the unmixed concentrations of the basic components in comparison to the

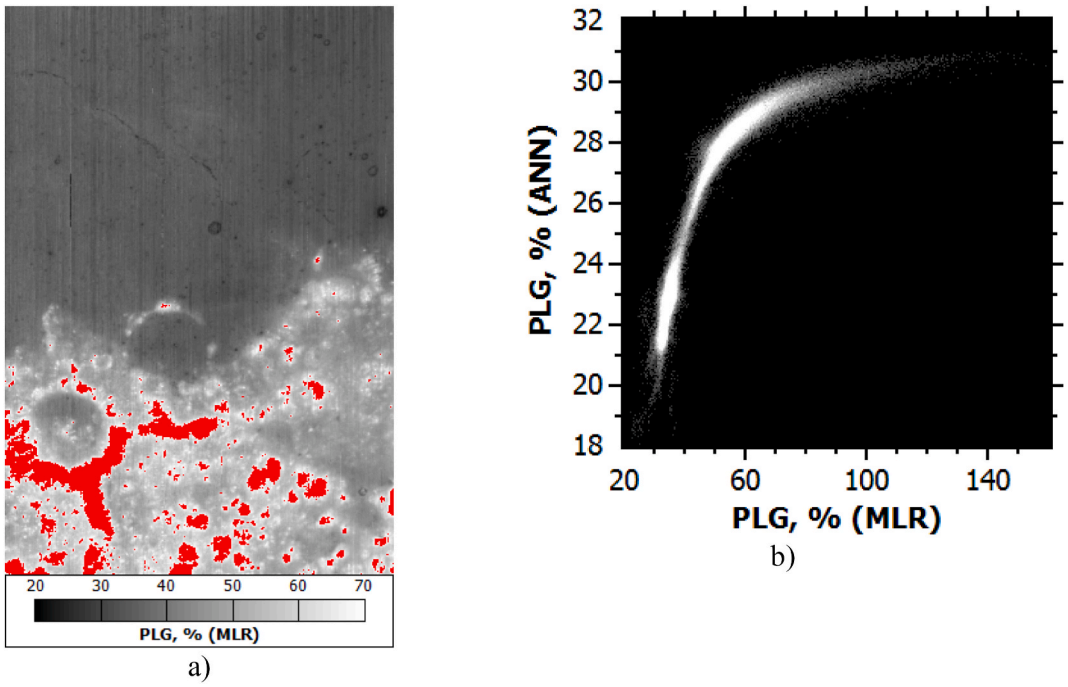

Fig. 26. (a) abundance of plagioclase calculated using Multiple Linear Regression (MLR) obtained from LSCC data (Eq. (3)); (b) correlation diagram "PLG, % (ANN) – PLG, % (MLR)".

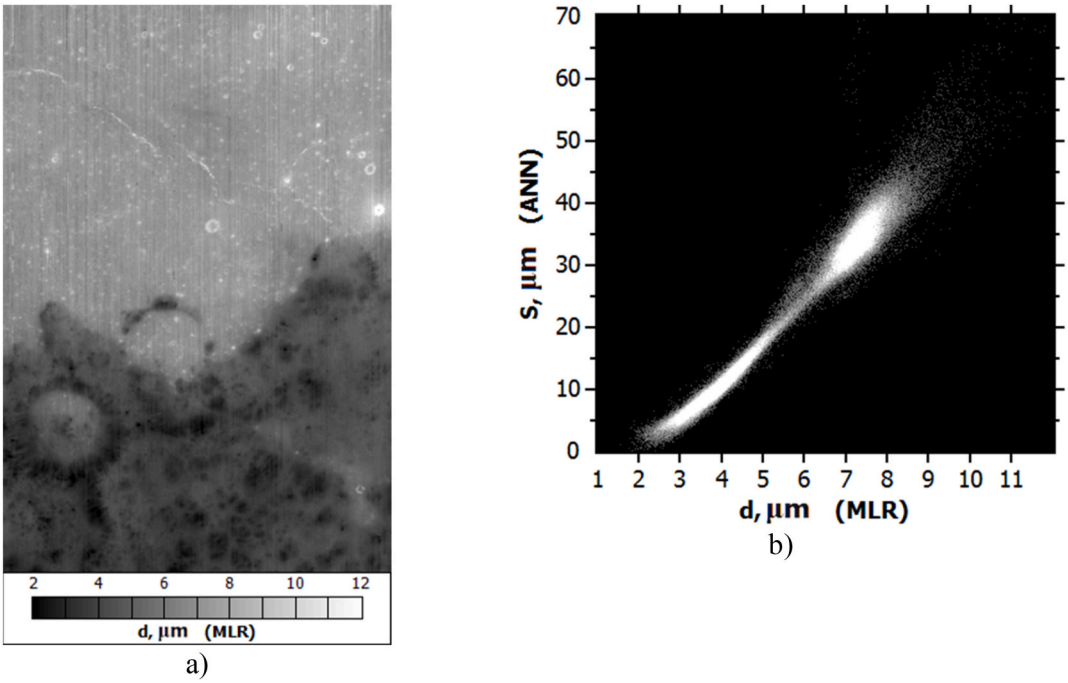

Fig. 27. (a) grain size calculated using Multiple Linear Regression (MLR) obtained from LSCC data (Eq. (4)); (b) correlation diagram "S (ANN) –d (MLR)".

measurements made in the laboratory by Taylor et al. (1996) (see the discussion in Sec. 3.2).

Thus, we can conclude that it is necessary to use the correlation relations which link the spectral and mineralogical parameters obtained from LSCC data with great caution, especially for highland areas. The significant difference in the distribution of the concentration values for the highlands (inverse correlation for maturity and clinopypoxene, and unrealistically high concentration values of plagioclase) is probably caused by the unrepresentativeness of the LSCC subsets used to find the MLR correlation for the highland areas.

Another possibility to determine the surface composition, physical and optical characteristics of the regolith by spectrophotometric observations is based on the investigation of the nonlinear correlations between laboratory measured optical and composition parameters of lunar LSCC samples using the correlation diagram, e.g. A(750 nm) – C (950/750 nm) (Lucey et al., 1995, 1998, 2000a, b, 2004). The formula for the so-called optical maturity (parameter OMATFe) was also obtained using this approach:

$$OMAT_{Fe} = \sqrt{(A(750\text{nm}) - 0.08)^2 + \left(\frac{A(950\text{nm})}{A(750\text{nm})} - 1.19\right)^2} \quad (6)$$

where $A(\lambda)$ is albedo. The parameter OMATFe is closely related to the maturity degree Is.

Fig. 28a displays the optical maturity map ($OMAT_{Fe}$) which was calculated by using Eq. (6). Comparison with our maturity parameter c_{nFe} (Fig. 19b) indicates a fairly high inverse correlation (Fig. 28b). As in Fig. 22b, regions for mare and highlands are well separated in the correlation diagram "c_{nFe} (ANN) – $OMAT_{Fe}$ (Lucey).

From the correlation diagram A (750 nm) – C (415/750 nm), Lucey's method (Lucey et al. (1995, 1998, 2000a, b, 2004) allows an estimation of the TiO_2 concentration:

$$TiO_2[\%] = 3.71\left\{arctan\left(\frac{A(415\text{nm})/A(750\text{nm}) - 0.42}{A(750\text{nm})}\right)\right\}^{5.98} \quad (7)$$

As ilmenite ($FeTiO_3$) (for example (Taylor, 1994),) contains most of the TiO_2 in the lunar regolith we want to compare the distribution of TiO_2 abundance calculated by Luceys's formula (7) (Fig. 29a) with the concentration of ilmenite c_{ILM} obtained by our method (Fig. 19h). For the highland areas, these parameters are well correlated ($c = 0.87$) with each other (Fig. 29b), although the nature of this correlation is nonlinear. For mare, the correlation is not observed. We want to point out that Lucey's formula (7) yields for this area a TiO_2 value which is too high for the lunar soil (up to 15%), indicating to use this approach with uttermost care.

We also compared the results of our new method with the results of a nonlinear approach proposed earlier by Korokhin et al. (2008), which applies machine learning, but with LSCC data, to find a correlation between compositional and optical parameters. Fig. 30a shows the TiO_2 abundance obtained by using a moderately nonlinear ANN (Korokhin et al., 2008). This map shows more realistic TiO_2 abundance values compared with the ones found through Lucey's approach (Fig. 29a), although both maps demonstrate a similar distribution of TiO_2 over the studied area, and the correlation diagrams have a similar shape. It should be noted that for highland areas, the approach of Korokhin et al. (2008) gives a more linear correlation between the content of ilmenite and TiO_2. Therefore, as expected, the correlation between the concentration of ilmenite c_{ILM} obtained using our new method (Fig. 19h) and TiO_2 abundance (Fig. 30a) is higher and equal to 0.91 (Fig. 30b).

Besides correlation methods, there are physical-based methods to derive the concentrations of basic minerals from spectral observations, for instance, from the analysis of the strength of the absorption band. Such an approach is described in Surkov et al. (2020), where a method of ilmenite abundance mapping from the distribution of the depth of the absorption band at 1550 nm has been proposed. We have therefore a possibility to carry out a direct comparison of our of ilmenite distribution (Fig. 19h) with the distribution (Fig. 31a) obtained using the method of Surkov et al. (2020). In general, both maps demonstrate the presence of common characteristic features of the distribution of ilmenite associated with morphological details on the lunar surface, for instance, the border between Mare Crisium and the highland, Crater Shapley, small young craters and rifts in Mare Crisium. However, our method gives twice the ilmenite content (up to 18%) compared with the approach of Surkov et al. (2020). The correlation diagram (Fig. 31c) expectedly has a two-branch structure. Ilmenite concentrations determined in mare areas show a more linear correlation ($c = 0.54$). Highland areas have no correlation. Such a difference can be explained by the difference in the used models. The concentration of ilmenite in Surkov et al., 2020b is obtained from the distribution of the depth of the absorption band at 1550 nm under the assumption that the band is formed only by ilmenite. But in this spectral range, there are spectral features associated with another basic mineral, first of all, clinopyroxene, which has a high albedo (see Fig. 9). In contrast to this, our approach is more general and takes into account the influence of all basic elements on the spectra and the nonlinear nature of the relation between compositional and spectral characteristics of the lunar soils, giving a different distribution of ilmenite compared to that obtained by Surkov et al. (2020).

6. Conclusions

1. A new method has been developed, combining the Shkuratov et al. (1999) model of nonlinear spectral mixing with machine learning for mapping the composition parameters of the lunar regolith from spectral observations.
2. The developed method performs order of magnitudes faster than the numerical optimization. This allows one to carry out a mapping the

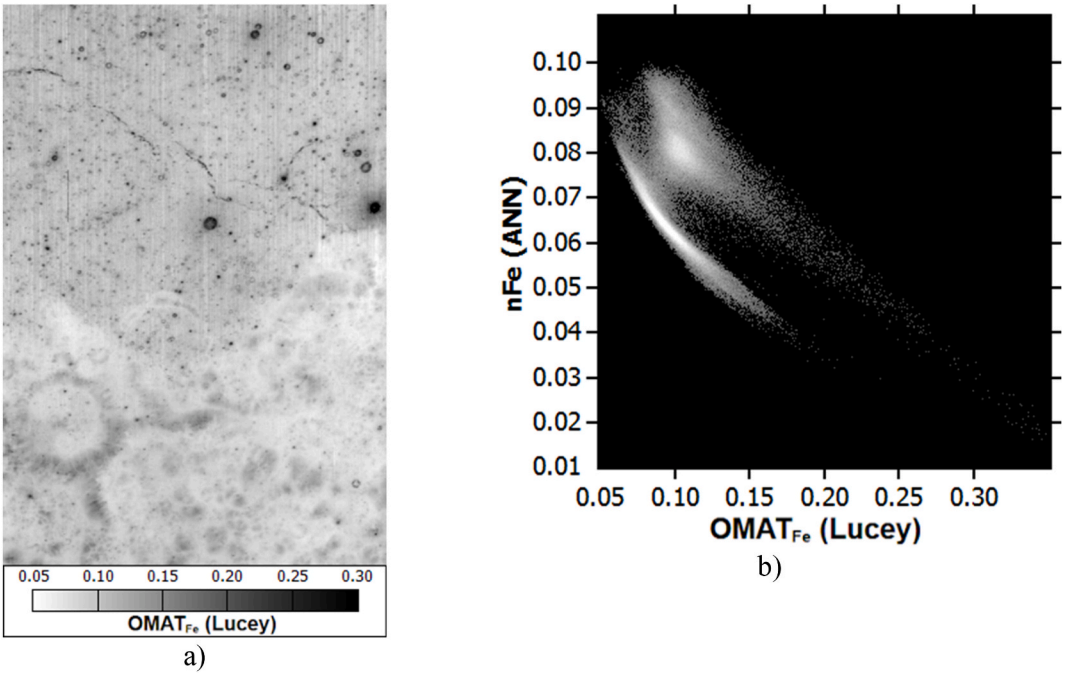

Fig. 28. (a) optical maturity $OMAT_{Fe}$ (negative image) by Lucey et al. (2000b) (Eq. (6)); (b) correlation diagram "c_{nFe} (ANN) – $OMAT_{Fe}$ (Lucey)".

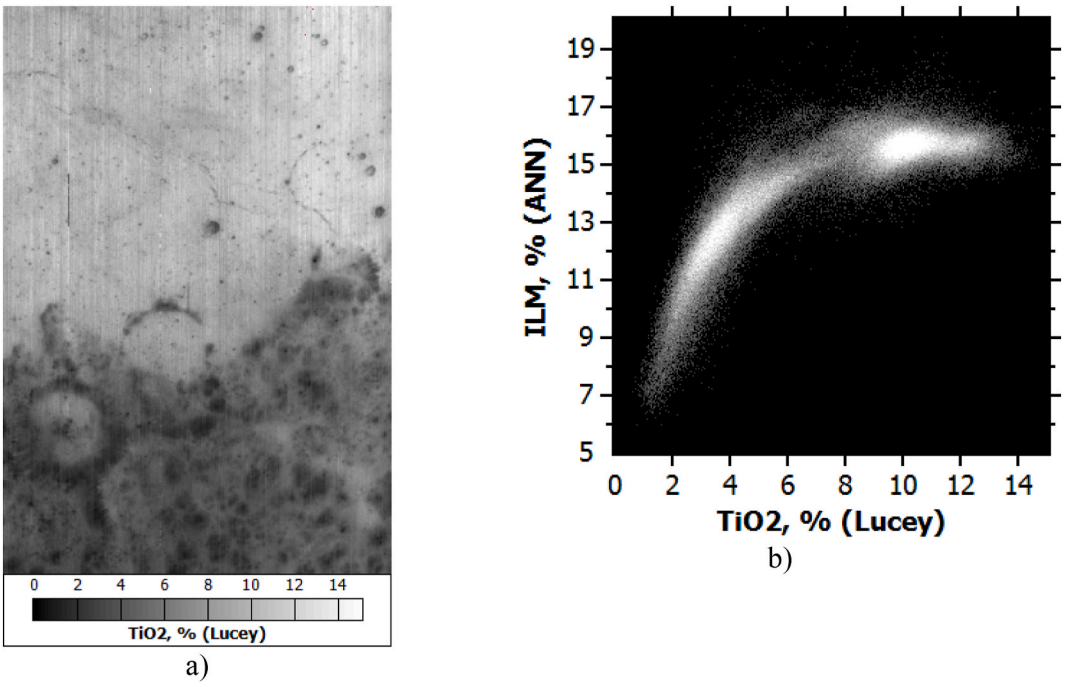

Fig. 29. (a) TiO_2 abundance determined by Lucey et al. (2000b) (Eq. (7)); (b) correlation diagram "ILM, % (ANN) – TiO_2, % (Lucey)".

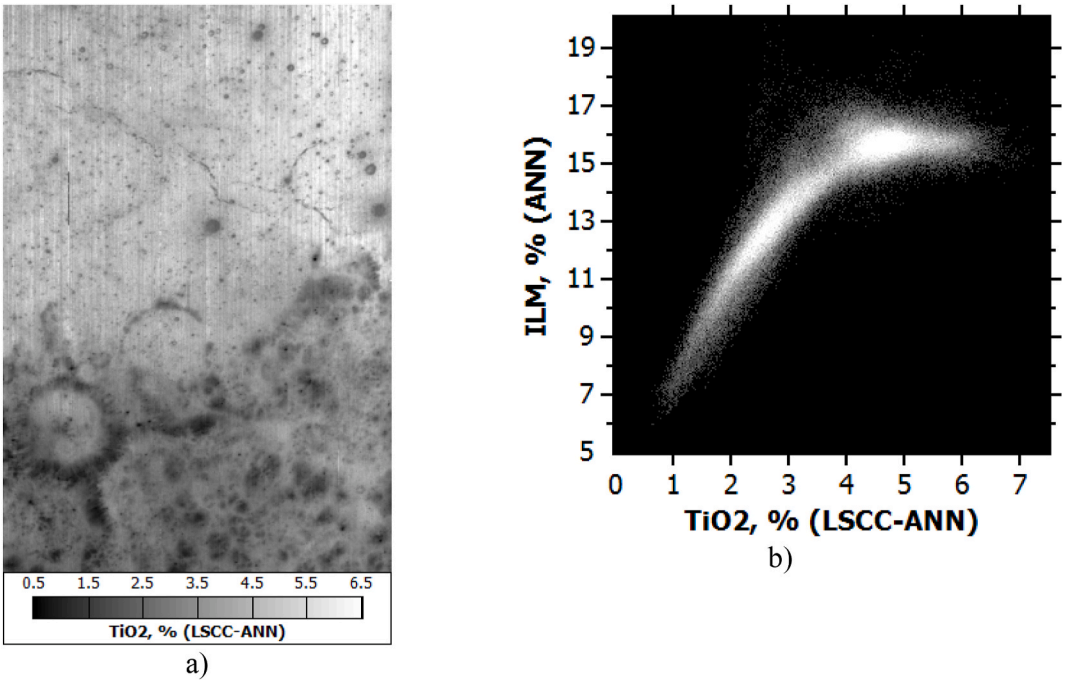

Fig. 30. (a) TiO_2 abundance obtained from non-linear regression derived from LSCC data using moderately nonlinear ANN (Korokhin et al., 2008); (b) correlation diagram "c_{ILM} (ANN) – TiO_2, % (LSCC-ANN)".

soil properties (composition, grain size and maturity) over large areas of the lunar surface.

3. A new set of basic minerals of the lunar soil is proposed, which can be used to model with the nonlinear mixing model (Shkuratov et al., 1999) adequately both the mare and highland spectra observed with the Chandrayaan-1 M^3 instrument (errors are not greater than 1.0–1.5%).
4. The set includes a virtual "gray component" with a "flat" (constant) spectrum accounting for the factors that change the general albedo, such as the spectrally neutral components (e.g., agglutinate glasses), errors in the photometric reduction and the incorrect value of porosity q, the real part of refractive index n and the grain size S for basic minerals.
5. A comparison with the results of other methods has been carried out, in particular with the popular multiple linear regression (MLR) approach (equations (3)–(5)). A high correlation is observed between the spectral mixing model parameter S and the grain size d ($c = 0.99$). We find high correlations between the ANN and the MLR-

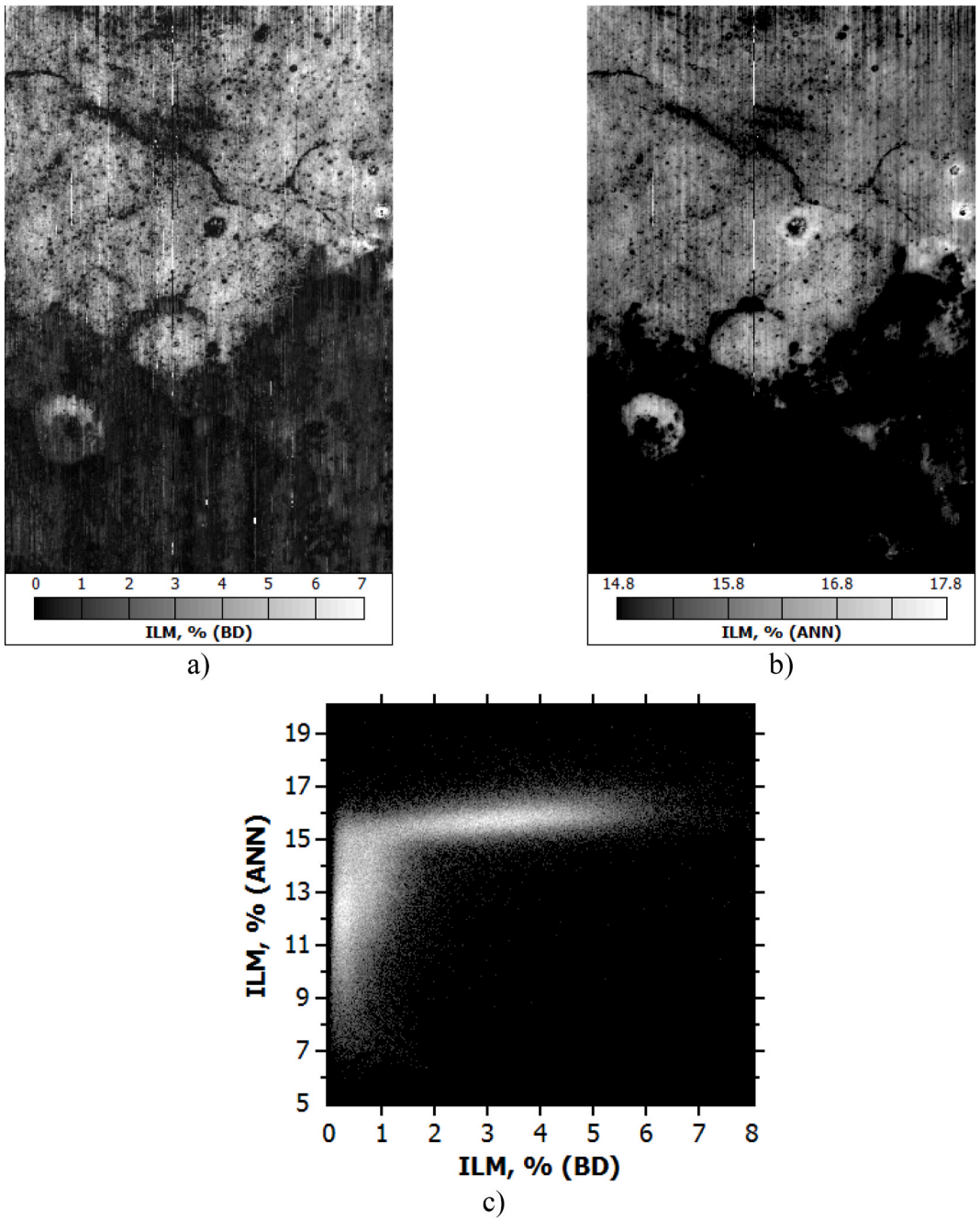

Fig. 31. Ilmenite abundance determined: (a) by the band depth (BD) (Surkov et al., 2020); (b) by our method (the same as in Fig. 19h, but with a different contrast). (c) Correlation diagram "ILM, % (ANN) – ILM, % (BD)".

concentrations of plagioclase in the mare areas ($c = 0.92$) and olivine in the highland areas ($c = 0.82$), between concentration of reduced nanophase iron c_{nFe} and the maturity parameter I_S ($c = 0.85$ for maria areas and $c = 0.83$ for highlands). The rest of the minerals either do not produce a direct correlation or even show an inverse correlation (clinopyroxene in the highlands). This can be explained by the fact that the assumption of the existence of linear dependence of concentration of these components on spectral parameters for deriving equations (3)–(5) do not hold.

6. We compared also our distribution of ilmenite with concentration of TiO_2 obtained by the Lucey's formula and the nonlinear approach proposed earlier by Korokhin et al. (2008). As expected, the second approach gives higher correlation with ilmenite (c = 0.91).
7. The proposed method allows to obtain geologically and physically reasonable parameters (concentration of basic minerals, grain size and maturity) of the lunar regolith due to the influence of all basic elements on spectra formation and the nonlinear nature of the relation between compositional and spectral characteristics of the lunar soils.
8. Future work will present results of detailed planetological studies using our new technique on specific target areas.
9. The method is suitable for the analysis of spectral observations not only of the lunar surface (e.g., data of Chandrayaan-1 M^3 and SIR-2); it can be easily extended to the analysis of other regolith-like surfaces of Solar system bodies.

CRediT authorship contribution statement

Viktor Korokhin: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yehor Surkov:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Urs Mall:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Vadym Kaydash:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Sergey Velichko:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Yuri Velikodsky:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Oksana Shalygina:** Writing – review & editing, Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix. Nonlinear spectral mixing model

The model is described in detail in Shkuratov et al. (1999). We summarize here only the basic formulas, which are used in this work.

A.1. Calculation of extinction coefficients spectra k(λ) from bidirectional reflectance spectra A(λ)

Since the diagnostic features of lunar regolith material in the visible and NIR spectral range occur due to multiply reflection of the light in the uppermost layer of the regolith's particulate medium, it is necessary to transform bidirectional reflectance spectra $A(\lambda)$ into absorption spectra $k(\lambda)$ which should be used for a correct spectral mixing. Shkuratov et al. (1999) provides a specially designed model which performs such a transformation.

The principal formulation of this model is

$$k(\lambda) = \tau\ \lambda\ /\ 4\ \pi\ S, \tag{A1}$$

where S is the average length of light propagation in a particle between internal reflections. In the first approximation, we can assume that this value is equal to the average particle grain size (see, for instance, (Pieters et al., 2002)). For simplicity, we call this parameter "grain size". The optical density τ (optical thickness between two consequent points of the internal reflection) is calculated as

$$\tau = -\ln\left[b \Big/ a + \sqrt{(b/a)^2 - c/a}\right] \tag{A2}$$

where

$a = T_e\ T_i\ (y\ R_i + q\ T_e)$,
$b = y\ R_b\ R_i + q\ T_e^2\ (1 + T_i)/2 - T_e\ (1 - q\ R_b)$,
$c = 2\ y\ R_b - 2\ T_e\ (1 - q\ R_b) + q\ T_e^2$,
$y = (1 - A(\lambda))^{\ 2}/2\ A(\lambda)$,

q is the volume fraction filled by the particles (packing density) in the medium (on surface), T_e and T_i are average transmittances for light penetrating into a particle from outside and in the opposite direction, R_b is the average backward reflectance coefficient, and R_i is the average coefficient of internal reflection inside the particle.

Let us assume that in visual and NIR light, the range of the real part of the refractive index is given by n (1.4 … 2.0, depending on the transparency of the material). Average reflectance coefficients can be calculated using the following empirical approximations:

$$R_e = r_0 + 0.055, \tag{A3}$$

$$R_b = (0.28\ n - 0.20)\ R_e, \tag{A4}$$

$$R_i = 1.04\text{–}1\ /\ n^2, \tag{A5}$$

where $r_0 = (n - 1)^{\ 2}/(n + 1)^{\ 2}$ being the Fresnel reflection coefficient at the normal incidence.

Average transmittances are calculated by

$$T_e = 1 - R_e, \tag{A6}$$

$$T_i = 1 - R_i. \quad \text{(A7)}$$

Using formula (A1), we can calculate the absorption spectrum for each component of a modeled mixture.

Let us note here that a knowledge of the mean grain size of laboratory measured samples of the minerals' mixture components is very important due to its strong influence on the albedo of the particulate surfaces. In contrast, the refractive index and the packing density have a weak effect on the albedo (see Fig. 3 from Shkuratov et al., 1999). Therefore, for simplicity in this work, we use $q = 0.5$ which is typical for lunar regolith, $n = 1.6$ for transparent components and $n = 2.0$ for nontransparent ones. A relatively low value of 2.0 is due to the limitations of equations (A2) – (A7) for wavelength <0.5 μm when the optical density τ for some nontransparent components, for instance, ilmenite is calculated.

A.2. Calculation of bidirectional reflectance spectra A(λ) of a surface from extinction coefficients spectra k(λ)

The inverse transformation $k(\lambda) \rightarrow A(\lambda)$ is found through the following steps.

To compute the albedo (reflectance) $A(\lambda)$ of a surface we need to compute the fractions of light flux scattered by a particle into the backward, r_b, and forward hemisphere, r_f.

r_b and r_f are calculated by using the following formulas:

$$r_b(\lambda) = R_b + H,\ r_f(\lambda) = R_f + T_e\ T_i \exp(-\tau) + H, \quad \text{(A8)}$$

where $H = 1/2\ T_e\ T_i\ R_i \exp(-2\tau)/(1 - R_i \exp(-\tau))$,

$R_f \cong R_e - R_b$ is average forward reflectance coefficient.

The optical density τ for a modeled medium (surface) consisting of transparent grains is calculated from its absorption spectrum using Eq. (A1):

$$\tau = k(\lambda)\ 4\ \pi\ S\ /\ \lambda. \quad \text{(A9)}$$

For a nontransparent medium (for instance, mineral ilmenite or agglutinate) τ from Eq. (A2) is used.

The next step is to use the indicatrix of a particle to calculate the albedo (reflectance) $A(\lambda)$ of a medium (surface). Denote q as the volume fraction filled by particles in a modeled medium. For a randomly packed medium, there is a mathematical theorem that the area fraction (or the length fraction) filled by particles in an intersecting plane (or a line) is equal to q. Thus, the one-dimensional indicatrix for a layer is

$$\rho_b = q\ r_b,$$

$$\rho_f = q\ r_f + 1 - q. \quad \text{(A10)}$$

Finally, the albedo of a surface (a half-infinite pile of the layers) may be calculated using the following formula

$$A(\lambda) = P - \sqrt{P^2 - 1} \quad \text{(A11)}$$

where $P = (1 + \rho_b^2 - \rho_f^2)/2\ \rho_b$.

A.3. Calculation of bidirectional reflectance spectra $A_j(\lambda)$ of a multicomponent surface

In Shkuratov's model, four types of surface structure are considered.

1) mixture of coarse particles (particle size $\gg \lambda$). They can be dark or transparent with different contribution of multiple scattering;
2) coarse transparent particles containing fine inclusions (inclusions size $\ll \lambda$) of an absorbing component. Such inclusions do not affect the refraction (n) of the particles but increase absorption (k), if their optical density is ≥ 1;
3) coarse particles surrounded by thin absorbing films (film thickness $\leq \lambda$). Such films change both refraction and absorption of the particle surfaces;
4) coarse transparent particles heterogeneous in scale $\ll \lambda$. Such particles can be considered as quasi-homogeneous with effective optical constants which should be calculated. Then the medium can be considered as one of type (1).

Practically the regolith of atmosphereless Solar System bodies can be considered as a combination of the structures mentioned above. Light scattering by each of them can be described by some generalization of the basic model.

Microscopy studies of the lunar samples indicate that for the lunar regolith, mixture type (1) is most likely applicable. In this case, the Shkuratov's model provides the following formulas for calculation of the one-dimensional indicatrix for a layer

$$\rho_b = q \sum (c(j)\ r_b(j)),$$

$$\rho_f = q \sum (c(j)\ r_f(j)) + 1 - q, \quad \text{(A12)}$$

where summation is performed by index of the component j, $c(j)$ is the concentration of component j. The albedo of a particle r_b and r_f are calculated using Eq. (A8).

Substituting Eq. (A12) in Eq. (A11) we obtain the albedo spectrum for a mixture of coarse particles.

A.4. The influence of space-weathering phenomena

A realistic modeling of measured lunar soil spectra requires one to consider the influence of space weathering on lunar surface materials. As is well

known (Starukhina and Shkuratov, 2001; Noble et al., 2007; Shkuratov et al., 2011), space-weathering phenomena include not only agglutinate formation. Another process is the formation of spherical grains of reduced iron of typical diameters 30–50 nm (up to ~300 nm), so-called nanophase iron (nFe^0). It most significantly changes the optical and spectral properties of lunar regolith. Although the volume fraction c_{Fe} of such grains is very small ($\sim 10^{-5} - 10^{-3}$), a high light absorption by these semitransparent inclusions results in a substantial darkening, spectral reddening, and a weakening of the absorption bands of the lunar soil spectra. nFe^0 grains are usually concentrated near the particle surfaces. In agglutinates, however, nFe^0 and submicron Fe^0 grains are dispersed all over the particle volume.

Let us assume, following Shkuratov et al. (1999), that soil particles are surrounded by a layer (rind) with a thickness $d = 100$ nm, containing grains of colloid iron. An idea to replace the spatial distribution of nFe^0 grains by their location in the particles rinds is physically preferable and is confirmed by direct microscopic investigations. Presence of nFe^0 grains in the rinds can be taken into account by the addition of the contribution of nFe^0 grains $\tau_{Fe}(\lambda)$ to the optical density of the soil particles in Eqs. (A8 – A9):

$$\tau_{mat}(\lambda) = \tau(\lambda) + \tau_{Fe}(\lambda) \quad \text{(A13)}$$

Where index "mat" stands for mature regolith particles as distinct from the immature ones.

The rind may be considered as a medium of type (2) from Sec. A.3, where the nFe^0 grains are distributed all over the rind volumes. In this case, the contribution of absorbing inclusions can be calculated using the Mie theory (Shkuratov et al., 1999):

$$\tau_{Fe}(\lambda) = 4\pi k_{Fe} d/\lambda = 6\pi(n/\lambda)\mathrm{Im}\{[(\check{n}_{Fe}/n)^2 - 1]/[(\check{n}_{Fe}/n)^2 + 2]\} c_{nFe}\, d, \quad \text{(A14)}$$

where $\check{n}_{Fe}$ is complex refractive index of iron, k_{Fe} is the spectrum of the effective value for the imaginary part of the refractive index of a silicate matrix with the refractive index n containing nFe^0-inclusions; c_{nFe} is the volume fraction of such inclusions, d is the size (thickness) of the medium. In our case, the size d in Eq. (A14) is equal to doubled rind thickness h due to the double intersection of the particle rims at each scattering of a light beam: $d = 2h \approx$ 200 nm.

As shown in Eq. (A14), $\tau_{Fe}(\lambda)$ depends only on the product of the reduced nanophase iron volume fraction c_{nFe} and the total size of iron-containing layer (d or $2h$). Fe-addition to τ can be taken in the form of Eq. (A14), where c_{nFe} is the effective average concentration of nFe^0 grains over the whole particle measured in ferromagnetic resonance analysis Morris (1977, 1980).

Eq. (A14) can be rewritten in the form:

$$\tau_{Fe}(\lambda) = \alpha_{Fe}(\lambda)\, c_{nFe}\, d, \quad \text{(A15)}$$

where $\alpha_{Fe}(\lambda)$ is the addition to the absorption coefficient $\alpha(\lambda) = 4\pi k/\lambda = \tau(\lambda)/d$ of a silicate matrix due to presence of nFe^0-inclusions ($\alpha_{Fe}(\lambda)$ being nFe^0 absorption per unit volume concentration of the inclusions).

Taking the optical constants of iron (compiled from Johnson and Cristy (1974) and Weaver et al. (1979)), we can tabulate the values of $\alpha_{Fe}(\lambda)$ (see Fig. A1), so the optical density $\tau_{mat}(\lambda)$ of the lunar particles can be calculated from that of the terrestrial analogs as a sum:

$$\tau_{mat}(\lambda) = \tau(\lambda) + \alpha_{Fe}(\lambda)\, c_{nFe}\, d. \quad \text{(A16)}$$

The optical constants of metals are sensitive to their structure, so the optical constants of nFe^0 may somewhat differ from those measured for terrestrial samples with a few kinds of imperfections and impurity. However, the effect of this difference is not expected to exceed the precision of the spectral model.

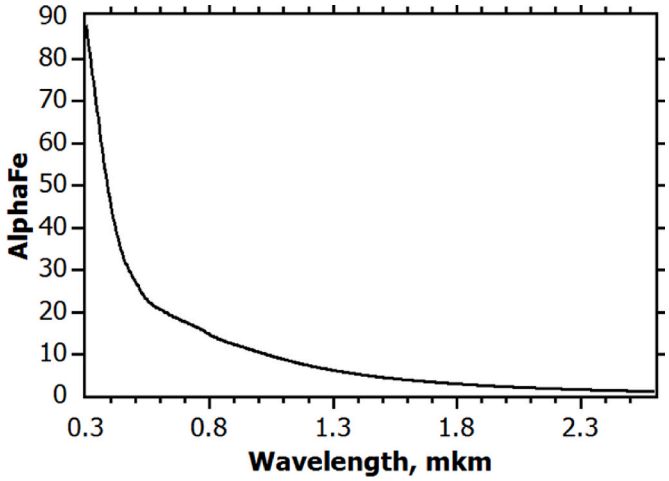

Fig. A1. α_{Fe} spectrum.

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